

Lecture 10

2026-04-30

Activity 1 (Always Anonymous)

QR Code or link

This is anonymous & optional, but helpful to Stats 20,



or here is a [direct link](#)

THERE IS NO MIDTERM

But let's review and plug some gaps

Extracting Information from Data Structures

Subsetting, using [- Data Frame

- The single bracket works on any data structure. The key is understanding what it is being applied to (input) and what is the result (output).
- This example involves a **data frame**, recall `df[row(s), column(s)]` will extract specific rows and columns if given them by name or by position (number or index)

```
EQ2025 <- read.table("https://service.scedc.caltech.edu/ftp/catalogs/SCEC_DC/2025.catalog",
                     col.names = c("DATE", "TIME", "EVENT_TYPE", "GT", "MAG", "MAGTYPE", "LAT",
                                   "LON", "DEPTH", "QUALITY", "EVENT_ID", "Num_Phases",
                                   "Num_Grams"))
str(EQ2025)
```

```
'data.frame':  16818 obs. of  13 variables:
 $ DATE      : chr  "2025/01/01" "2025/01/01" "2025/01/01" "2025/01/01" ...
 $ TIME      : chr  "01:00:56.92" "01:16:28.99" "01:30:49.87" "01:54:51.66" ...
 $ EVENT_TYPE: chr  "eq" "eq" "eq" "eq" ...
 $ GT        : chr  "1" "1" "1" "1" ...
 $ MAG       : num  0.59 1.21 0.6 1.06 0.57 0.96 0.9 0.58 0.85 0.51 ...
 $ MAGTYPE   : chr  "1" "1" "1" "1" ...
 $ LAT       : num  33.3 34 34.1 35.5 33.5 ...
 $ LON       : num  -116 -117 -118 -118 -117 ...
```

```
$ DEPTH      : num  3.3 13.4 12.2 8.6 5.9 10.2 8.8 7.9 4.9 10.9 ...
$ QUALITY    : chr   "A" "A" "A" "A" ...
$ EVENT_ID   : int  40831087 40831095 40831103 40831111 40831119 40831127 40831135 40831151 40831159 40831167 ...
$ Num_Phases : int    7 94 18 49 58 96 74 41 35 44 ...
$ Num_Grams  : int  1005 2122 1834 1442 1125 1472 1194 1106 1557 1880 ...
```

Subsetting, using [- Specifying columns

- With a data frame, we could use it to extract single vectors or multiple vectors, the single vector result will be a vector, multiple vectors return a data frame.

```
mag1 <- EQ2025[ , 5] # column 5, all rows
mag2 <- EQ2025[ , "MAG"] # name of the column, all rows
paste("Compare mag1 and mag2", all.equal(mag1, mag2))
```

```
[1] "Compare mag1 and mag2 TRUE"
```

```
is(mag1)
```

```
[1] "numeric" "vector"
```

```
mag_qual1 <- EQ2025[ , c(5, 10)] # columns 5 and 10, all rows
mag_qual2 <- EQ2025[ , c("MAG", "QUALITY")]
mag_qual3 <- EQ2025[ , c("QUALITY", "MAG")] # switched, you can order things the way you want
paste("Are mag_qual1 and mag_qual2 the same?", all.equal(mag_qual1, mag_qual2))
```

```
[1] "Are mag_qual1 and mag_qual2 the same? TRUE"
```

```
paste("Are mag_qual2 and mag_qual3 the same?", all.equal(mag_qual2, mag_qual3))
```

```
[1] "Are mag_qual2 and mag_qual3 the same? Names: 2 string mismatches"
[2] "Are mag_qual2 and mag_qual3 the same? Component 1: Modes: numeric, character"
[3] "Are mag_qual2 and mag_qual3 the same? Component 1: target is numeric, current is character"
[4] "Are mag_qual2 and mag_qual3 the same? Component 2: Modes: character, numeric"
[5] "Are mag_qual2 and mag_qual3 the same? Component 2: target is character, current is numeric"
```

```
is(mag_qual2)
```

```
[1] "data.frame" "list"          "oldClass"    "vector"
```

Subsetting, using [- Specifying rows

- Of course you can always use row indices BUT we are at the point where you may wish to subset based on characteristics of the row instead of something like 1:10.

```
mag3 <- EQ2025[EQ2025$MAG > 3, ] # The row values for the MAG column must be > 3, all columns
mag3safer <- EQ2025[which(EQ2025$MAG > 3), ] # name of the column, all rows
all.equal(mag3, mag3safer) # result is the same
```

```
[1] TRUE
```

```
which(EQ2025$MAG > 3)
```

```

[1] 273 367 475 557 678 878 902 1624 1636 1656 1746 1777
[13] 1784 1793 1877 2040 2134 2342 2432 2565 2595 2724 2849 2986
[25] 3391 3453 3475 3839 3902 3911 3969 4172 4231 4243 4343 4669
[37] 4799 5248 5395 5670 6180 6608 6899 7200 7462 8602 8792 8793
[49] 8826 8831 8858 9065 9286 9550 9865 9866 9992 9993 10120 10239
[61] 10255 10274 10310 10328 10329 10333 10336 10528 10750 10788 10809 11679
[73] 11943 12065 12121 12160 12295 12358 12362 12556 13054 13145 13203 13214
[85] 13220 13932 13961 13997 14073 14082 14314 14794 14849 14879 14892 15144
[97] 15186 15191 15192 15447 15648 15698 15823 15938 16076 16094 16105 16118
[109] 16181 16226

```

- The result of `EQ2025$MAG > 3` gives us a vector of TRUE and FALSE for all rows
- `which()` will give us the exact row indices, it can be safer when you are using `==` or have NA, NaN values

Using the \$

- We can use it to extract vectors by their names (must be named)
- It is a little more readable but the tradeoff is more limitations

```

MAG <- EQ2025$MAG # creates a vector unassociated with the data frame, free floating, can be confusing

## a better application
fivenum(EQ2025$MAG) # extracting a specific vector for analysis

```

```

[1] 0.00 0.71 0.99 1.34 5.21

```

```

## you could extract further
fivenum(EQ2025$MAG[EQ2025$MAG > 3]) # vector EQ2025$MAG extract those MAG > 3

```

```

[1] 3.01 3.12 3.29 3.57 5.21

```

```
## and use different information
```

```
table(EQ2025$QUALITY[EQ2025$MAG > 3]) # vector EQ2025$QUALITY extract those with MAG > 3
```

	A	B	C	D
103	1	4	2	

Subsetting a subset

This can look daunting, we are taking a subset of one variable given the values on a different variable in the last example on the previous slide.

```
table(EQ2025$QUALITY[EQ2025$MAG > 3])
```

```
# compare EQ2025[EQ2025$MAG > 3, ] from above or EQ2025[EQ2025$MAG > 3, "QUALITY" ] or EQ2025[EQ2025$MAG > 3, 10 ]
```

This is where the “then” operator (`|>`) helps, we extract the exact subset we want and then `table()` the subset. The code is easier to read.

```
EQ2025$QUALITY[EQ2025$MAG > 3] |> table()
```

- R does not look inside your data frame, you must explicitly reference columns (e.g., `EQ2025$MAG`) on the row side of `[r, c]`
- The second argument (columns) must be column indices or names, not the actual values of a column.

Activity #2 Try it

Try some subsetting and ask questions

```
# how do you calculate a five number summary for magnitude (MAG) given a depth (DEPTH) greater than 12km?
```

If your median (3rd value) was 0.89 that is correct.

Subsetting, using `[[`

- `[[ ]]` extracts **exactly one element**
- and returns its contents (not the container/not the outer structure)
- uses position or names (if available).

```
rivers[[141]] # not the best use but an example
```

```
[1] 1770
```

```
iris1N <- iris[["Sepal.Length"]] # a lot of typing for one column...  
iris1P <- iris[[1]]  
all.equal(iris1N, iris1P)
```

```
[1] TRUE
```

```
is(iris1P)
```

```
[1] "numeric" "vector"
```

- IMO the best application is with lists

List Example

- Extract the “pivot” (the column order used in model estimation for numerical stability)

lm()

```
modell1 <- lm(Fertility ~ Education, data = swiss)
str(modell1)
```

List of 12

```
$ coefficients : Named num [1:2] 79.61 -0.862
..- attr(*, "names")= chr [1:2] "(Intercept)" "Education"
$ residuals    : Named num [1:47] 10.9 11.3 17.2 12.2 10.2 ...
..- attr(*, "names")= chr [1:47] "Courtelary" "Delemont" "Franches-Mnt" "Moutier" ...
$ effects      : Named num [1:47] -480.9 -56.2 14.9 10.2 9.4 ...
..- attr(*, "names")= chr [1:47] "(Intercept)" "Education" "" "" ...
$ rank         : int 2
$ fitted.values: Named num [1:47] 69.3 71.8 75.3 73.6 66.7 ...
..- attr(*, "names")= chr [1:47] "Courtelary" "Delemont" "Franches-Mnt" "Moutier" ...
$ assign       : int [1:2] 0 1
$ qr           :List of 5
..$ qr        : num [1:47, 1:2] -6.856 0.146 0.146 0.146 0.146 ...
.. ..- attr(*, "dimnames")=List of 2
.. .. ..$ : chr [1:47] "Courtelary" "Delemont" "Franches-Mnt" "Moutier" ...
.. .. ..$ : chr [1:2] "(Intercept)" "Education"
.. ..- attr(*, "assign")= int [1:2] 0 1
..$ qraux: num [1:2] 1.15 1.03
..$ pivot: int [1:2] 1 2
..$ tol   : num 1e-07
..$ rank  : int 2
```



```

..- attr(*, "class")= chr "qr"
$ df.residual : int 45
$ xlevels : Named list()
$ call : language lm(formula = Fertility ~ Education, data = swiss)
$ terms :Classes 'terms', 'formula' language Fertility ~ Education
.. ..- attr(*, "variables")= language list(Fertility, Education)
.. ..- attr(*, "factors")= int [1:2, 1] 0 1
.. ..- attr(*, "dimnames")=List of 2
.. ..$ : chr [1:2] "Fertility" "Education"
.. ..$ : chr "Education"
.. ..- attr(*, "term.labels")= chr "Education"
.. ..- attr(*, "order")= int 1
.. ..- attr(*, "intercept")= int 1
.. ..- attr(*, "response")= int 1
.. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
.. ..- attr(*, "predvars")= language list(Fertility, Education)
.. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
.. ..- attr(*, "names")= chr [1:2] "Fertility" "Education"
$ model :'data.frame': 47 obs. of 2 variables:
..$ Fertility: num [1:47] 80.2 83.1 92.5 85.8 76.9 76.1 83.8 92.4 82.4 82.9 ...
..$ Education: int [1:47] 12 9 5 7 15 7 7 8 7 13 ...
..- attr(*, "terms")=Classes 'terms', 'formula' language Fertility ~ Education
.. ..- attr(*, "variables")= language list(Fertility, Education)
.. ..- attr(*, "factors")= int [1:2, 1] 0 1
.. ..- attr(*, "dimnames")=List of 2
.. ..$ : chr [1:2] "Fertility" "Education"
.. ..$ : chr "Education"
.. ..- attr(*, "term.labels")= chr "Education"
.. ..- attr(*, "order")= int 1
.. ..- attr(*, "intercept")= int 1
.. ..- attr(*, "response")= int 1

```

```

.. .. ..- attr(*, ".Environment")=<environment: R_GlobalEnv>
.. .. ..- attr(*, "predvars")= language list(Fertility, Education)
.. .. ..- attr(*, "dataClasses")= Named chr [1:2] "numeric" "numeric"
.. .. ..- attr(*, "names")= chr [1:2] "Fertility" "Education"
- attr(*, "class")= chr "lm"

```

extract

```
names(model1)
```

```

[1] "coefficients" "residuals"      "effects"        "rank"
[5] "fitted.values" "assign"          "qr"             "df.residual"
[9] "xlevels"      "call"           "terms"         "model"

```

```
model1[["qr"]][["pivot"]] # notice, it extracts one component from many
```

```
[1] 1 2
```

```
model1$qr$pivot # this works too though
```

```
[1] 1 2
```

You might ask why...

```

library(jsonlite)
junk <- read_json("http://www.stat.ucla.edu/~vlew/datasets/DataSift-49de7967b29ac130ab49248db140d634-1389311635.json")
names(junk)

```

```
[1] "count"      "hash"      "hash_type" "id"        "delivered_at"
[6] "interactions"
```

```
str(junk$interactions)
```

```
List of 18
```

```
$ :List of 6
..$ interaction:List of 8
.. ..$ author :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/1833467114/oxford_2_normal.jpg"
.. .. ..$ id : int 494925931
.. .. ..$ link : chr "http://twitter.com/OxfordFreegle"
.. .. ..$ name : chr "Oxford Freegle"
.. .. ..$ username: chr "OxfordFreegle"
.. ..$ content : chr "WANTED: Blue Brooches - Any style or condition (Abingdon Ox14) http://t.co/cETDGisx"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:04 +0000"
.. ..$ id : chr "1e25dd61585aa200e07405b0f7d6e7ec"
.. ..$ link : chr "http://twitter.com/OxfordFreegle/statuses/290596104419041280"
.. ..$ schema :List of 1
.. .. ..$ version: int 3
.. ..$ source : chr "Freegle"
.. ..$ type : chr "twitter"
..$ klout :List of 1
.. ..$ score: int 25
..$ language :List of 2
.. ..$ confidence: int 62
.. ..$ tag : chr "en"
..$ links :List of 8
.. ..$ code :List of 1
.. .. ..$ : int 200
.. ..$ created_at :List of 1
```

```

.. ..$ : chr "Sun, 13 Jan 2013 23:08:05 +0000"
.. ..$ hops :List of 1
.. ..$ : list()
.. ..$ meta :List of 3
.. ..$ charset :List of 1
.. ..$ : chr "UTF-8"
.. ..$ content_type:List of 1
.. ..$ : chr "text/html"
.. ..$ lang :List of 1
.. ..$ : chr "unknown"
.. ..$ normalized_url:List of 1
.. ..$ : chr "http://go.frgl.it/Z39rd"
.. ..$ retweet_count :List of 1
.. ..$ : int 0
.. ..$ title :List of 1
.. ..$ : NULL
.. ..$ url :List of 1
.. ..$ : chr "http://go.frgl.it/Z39rd"
..$ salience :List of 1
.. ..$ content:List of 1
.. ..$ sentiment: int 0
..$ twitter :List of 7
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:04 +0000"
.. ..$ domains :List of 1
.. ..$ : chr "go.frgl.it"
.. ..$ id : chr "290596104419041280"
.. ..$ links :List of 1
.. ..$ : chr "http://go.frgl.it/Z39rd"
.. ..$ source : chr "<a href=\"http://ilovefreegle.org\" rel=\"nofollow\">Freegle</a>"
.. ..$ text : chr "WANTED: Blue Brooches - Any style or condition (Abingdon Ox14) http://t.co/cETDGisx"
.. ..$ user :List of 15

```

```

.. .. ..$ created_at      : chr "Fri, 17 Feb 2012 12:08:40 +0000"
.. .. ..$ description     : chr "An online reuse community group, which aims to save stuff being thrown away which
.. .. ..$ followers_count : int 167
.. .. ..$ friends_count  : int 134
.. .. ..$ id              : int 494925931
.. .. ..$ id_str          : chr "494925931"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 1
.. .. ..$ location        : chr "Oxford, England"
.. .. ..$ name            : chr "Oxford Freegle"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/1833467114/oxford_2_normal.jpg"
.. .. ..$ screen_name     : chr "OxfordFreegle"
.. .. ..$ statuses_count  : int 11339
.. .. ..$ time_zone       : chr "London"
.. .. ..$ url             : chr "http://www.freegle.in/oxford"
$ :List of 6
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/2845454798/beafdff7258a4f6965f309779c4882cb_normal.jpeg"
.. .. ..$ id       : int 309914105
.. .. ..$ link     : chr "http://twitter.com/ImRelevance"
.. .. ..$ name     : chr "Relevance"
.. .. ..$ username: chr "ImRelevance"
.. ..$ content    : chr "I uploaded a @YouTube video http://t.co/X1aUIU7k Relevance - Player - Neyo Style Beat"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:12 +0000"
.. ..$ id         : chr "1e25dd61a4a5a600e074697e4d243d98"
.. ..$ link       : chr "http://twitter.com/ImRelevance/statuses/290596136564166656"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Google"
.. ..$ type       : chr "twitter"

```

```

..$ klout          :List of 1
.. ..$ score: int 29
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag          : chr "en"
..$ links           :List of 8
.. ..$ code         :List of 1
.. .. ..$ : int 200
.. ..$ created_at   :List of 1
.. .. ..$ : chr "Sun, 13 Jan 2013 23:08:13 +0000"
.. ..$ hops         :List of 1
.. .. ..$ :List of 1
.. .. .. ..$ : chr "http://youtu.be/a2_WZLhPQsg?a"
.. ..$ meta         :List of 6
.. .. ..$ charset   :List of 1
.. .. .. ..$ : chr "UTF-8"
.. .. ..$ content_type:List of 1
.. .. .. ..$ : chr "text/html"
.. .. ..$ keywords   :List of 1
.. .. .. ..$ :List of 3
.. .. .. .. ..$ : chr "relevance"
.. .. .. .. ..$ : chr "relevance beats"
.. .. .. .. ..$ : chr "produced by relevance"
.. .. ..$ lang       :List of 1
.. .. .. ..$ : chr "en"
.. .. ..$ opengraph   :List of 1
.. .. .. ..$ :List of 6
.. .. .. .. ..$ site_name : chr "YouTube"
.. .. .. .. ..$ description: chr ""
.. .. .. .. ..$ type       : chr "video"
.. .. .. .. ..$ url        : chr "http://www.youtube.com/watch?v=a2_WZLhPQsg"

```

```

.. ..$ title      : chr "Relevance - Player - Neyo Style Beat"
.. ..$ image      : chr "http://i2.ytimg.com/vi/a2_WZLhPQsg/mqdefault.jpg"
.. ..$ twitter    :List of 1
.. ..$ :List of 2
.. ..$ player_width : chr "1280"
.. ..$ player_height: chr "720"
.. ..$ normalized_url:List of 1
.. ..$ : chr "http://youtube.com/watch?v=a2_WZLhPQsg"
.. ..$ retweet_count :List of 1
.. ..$ : int 0
.. ..$ title      :List of 1
.. ..$ : chr "Relevance - Player - Neyo Style Beat - YouTube"
.. ..$ url        :List of 1
.. ..$ : chr "http://www.youtube.com/watch?v=a2_WZLhPQsg"
..$ salience    :List of 1
.. ..$ content:List of 1
.. ..$ sentiment: int 0
..$ twitter     :List of 9
.. ..$ created_at : chr "Sun, 13 Jan 2013 23:08:12 +0000"
.. ..$ domains    :List of 1
.. ..$ : chr "youtu.be"
.. ..$ id         : chr "290596136564166656"
.. ..$ links      :List of 1
.. ..$ : chr "http://youtu.be/a2_WZLhPQsg?a"
.. ..$ mention_ids:List of 1
.. ..$ : int 10228272
.. ..$ mentions   :List of 1
.. ..$ : chr "YouTube"
.. ..$ source     : chr "<a href=\"http://www.google.com/\" rel=\"nofollow\">Google</a>"
.. ..$ text       : chr "I uploaded a @YouTube video http://t.co/X1aUIU7k Relevance - Player - Neyo Style Beat"
.. ..$ user       :List of 16

```

```

.. .. ..$ created_at      : chr "Thu, 02 Jun 2011 21:52:21 +0000"
.. .. ..$ description     : chr "Producer / Emcee "
.. .. ..$ followers_count  : int 112
.. .. ..$ friends_count   : int 319
.. .. ..$ id              : int 309914105
.. .. ..$ id_str          : chr "309914105"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 1
.. .. ..$ location        : chr "Orlando, Florida"
.. .. ..$ name            : chr "Relevance"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2845454798/beafdff7258a4f6965f309779c4882cb_normal.jpg"
.. .. ..$ screen_name     : chr "ImRelevance"
.. .. ..$ statuses_count  : int 119
.. .. ..$ time_zone       : chr "Eastern Time (US & Canada)"
.. .. ..$ url             : chr "http://www.facebook.com/relevance"
.. .. ..$ utc_offset      : int -18000
$ :List of 5
..$ demographic:List of 1
.. ..$ gender: chr "male"
..$ interaction:List of 8
.. ..$ author :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/3089748450/6f24c96b7b973b606506e8a11016a137_normal.jpg"
.. .. ..$ id      : int 83495660
.. .. ..$ link    : chr "http://twitter.com/redboyman"
.. .. ..$ name    : chr "Aldo Ortega"
.. .. ..$ username: chr "redboyman"
.. ..$ content   : chr "Ya estamos en el 2013. Ya déjen de poner el puto Gangnam Style por favor!"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:13 +0000"
.. ..$ id        : chr "1e25dd61ae2eac80e074f8a63773f51c"
.. ..$ link      : chr "http://twitter.com/redboyman/statuses/290596140418740224"
.. ..$ schema    :List of 1

```



```

.. .. ..$ version: int 3
.. ..$ source      : chr "TweetDeck"
.. ..$ type        : chr "twitter"
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag         : chr "es"
..$ salience       :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 4
..$ twitter         :List of 5
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:13 +0000"
.. ..$ id          : chr "290596140418740224"
.. ..$ source      : chr "<a href=\"http://www.tweetdeck.com\" rel=\"nofollow\">TweetDeck</a>"
.. ..$ text        : chr "Ya estamos en el 2013. Ya déjen de poner el puto Gangnam Style por favor!"
.. ..$ user        :List of 14
.. .. ..$ created_at      : chr "Mon, 19 Oct 2009 01:42:05 +0000"
.. .. ..$ description     : chr "FRESH"
.. .. ..$ followers_count : int 188
.. .. ..$ friends_count   : int 176
.. .. ..$ id              : int 83495660
.. .. ..$ id_str          : chr "83495660"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 2
.. .. ..$ name            : chr "Aldo Ortega"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3089748450/6f24c96b7b973b606506e8a11016a137_normal.jpg"
.. .. ..$ screen_name     : chr "redboyman"
.. .. ..$ statuses_count  : int 3726
.. .. ..$ time_zone       : chr "Mountain Time (US & Canada)"
.. .. ..$ utc_offset      : int -25200
$ :List of 5
..$ demographic:List of 1

```

```

.. ..$ gender: chr "mostly_female"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/2535170667/2xbno5wguhrwd0yv82x_normal.jpeg"
.. .. ..$ id       : int 273765384
.. .. ..$ link     : chr "http://twitter.com/Bobbie_Austin"
.. .. ..$ name     : chr "The Notorious B.O.B"
.. .. ..$ username: chr "Bobbie_Austin"
.. ..$ content     : chr "Watching the Golden Globes red carpet on E! I'll definitely be tweeting my personal commenta
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:24 +0000"
.. ..$ id         : chr "1e25dd621716a400e074c3e735015bf8"
.. ..$ link       : chr "http://twitter.com/Bobbie_Austin/statuses/290596186711261186"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Echofon"
.. ..$ type       : chr "twitter"
..$ language     :List of 2
.. ..$ confidence: int 100
.. ..$ tag        : chr "en"
..$ salience    :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 12
..$ twitter     :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:24 +0000"
.. ..$ hashtags  :List of 1
.. .. ..$ : chr "ERedCarpet"
.. ..$ id        : chr "290596186711261186"
.. ..$ source    : chr "<a href=\"http://www.echofon.com/\" rel=\"nofollow\">Echofon</a>"
.. ..$ text      : chr "Watching the Golden Globes red carpet on E! I'll definitely be tweeting my personal commenta
.. ..$ user      :List of 16
.. .. ..$ created_at      : chr "Tue, 29 Mar 2011 03:37:04 +0000"

```

```

.. .. ..$ description      : chr "Bobbie Austin is a fictional character in which I play the role of a fashion craze
.. .. ..$ followers_count  : int 401
.. .. ..$ friends_count    : int 848
.. .. ..$ id               : int 273765384
.. .. ..$ id_str           : chr "273765384"
.. .. ..$ lang             : chr "en"
.. .. ..$ listed_count     : int 5
.. .. ..$ location         : chr "Jersey"
.. .. ..$ name             : chr "The Notorious B.O.B"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2535170667/2xbno5wguhrwd0yv82x_normal.jpeg"
.. .. ..$ screen_name      : chr "Bobbie_Austin"
.. .. ..$ statuses_count   : int 4366
.. .. ..$ time_zone        : chr "Quito"
.. .. ..$ url              : chr "http://www.bobbieaustinscloset.com/"
.. .. ..$ utc_offset       : int -18000
$:List of 6
..$ demographic:List of 1
.. ..$ gender: chr "female"
..$ interaction:List of 9
.. ..$ author :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/3101306838/a50e5650b00a9eaeb07f2234bf25ff3d_normal.jpeg"
.. .. ..$ id      : int 448078833
.. .. ..$ link    : chr "http://twitter.com/jelly_marieeee"
.. .. ..$ name    : chr "jelly lane "
.. .. ..$ username: chr "jelly_marieeee"
.. ..$ content   : chr "I alwaaayyss have to pick out my outfit the night before school. #girlproblems"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:26 +0000"
.. ..$ geo       :List of 2
.. .. ..$ latitude : num 43.3
.. .. ..$ longitude: num -71.5
.. ..$ id        : chr "1e25dd622a29a100e07400dbc65b3c92"

```

```

.. ..$ link      : chr "http://twitter.com/jelly_marieeee/statuses/290596193455710208"
.. ..$ schema    :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Twitter for iPhone"
.. ..$ type       : chr "twitter"
..$ klout         :List of 1
.. ..$ score: int 34
..$ language      :List of 2
.. ..$ confidence: int 100
.. ..$ tag         : chr "en"
..$ salience     :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter       :List of 8
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:26 +0000"
.. ..$ geo         :List of 2
.. .. ..$ latitude : num 43.3
.. .. ..$ longitude: num -71.5
.. ..$ hashtags   :List of 1
.. .. ..$ : chr "girlproblems"
.. ..$ id          : chr "290596193455710208"
.. ..$ place       :List of 7
.. .. ..$ country   : chr "United States"
.. .. ..$ country_code: chr "US"
.. .. ..$ full_name  : chr "Loudon, NH"
.. .. ..$ id        : chr "6f4c572ccc46d8da"
.. .. ..$ name       : chr "Loudon"
.. .. ..$ place_type : chr "city"
.. .. ..$ url        : chr "http://api.twitter.com/1/geo/id/6f4c572ccc46d8da.json"
.. ..$ source      : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. ..$ text        : chr "I alwaaayyss have to pick out my outfit the night before school. #girlproblems"

```

```

.. ..$ user      :List of 14
.. .. ..$ created_at      : chr "Tue, 27 Dec 2011 16:01:25 +0000"
.. .. ..$ description     : chr "Be the change you want to see in the world."
.. .. ..$ followers_count : int 267
.. .. ..$ friends_count   : int 131
.. .. ..$ geo_enabled     : logi TRUE
.. .. ..$ id              : int 448078833
.. .. ..$ id_str          : chr "448078833"
.. .. ..$ lang            : chr "en"
.. .. ..$ name            : chr "jelly lane "
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3101306838/a50e5650b00a9eaeb07f2234bf25ff3d_normal.jpg"
.. .. ..$ screen_name     : chr "jelly_marieeee"
.. .. ..$ statuses_count  : int 6860
.. .. ..$ time_zone       : chr "Eastern Time (US & Canada)"
.. .. ..$ utc_offset      : int -18000
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "male"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/3104325484/a158d36a6aae44df6fe6ad791b9d7c47_normal.jpeg"
.. .. ..$ id       : int 877378586
.. .. ..$ link     : chr "http://twitter.com/geOrga_"
.. .. ..$ name     : chr "bruce wayne"
.. .. ..$ username: chr "geOrga_"
.. ..$ content    : chr "so sick of how i look and what i dress like"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:29 +0000"
.. ..$ id         : chr "1e25dd6246c5a480e074a00e37861ee8"
.. ..$ link       : chr "http://twitter.com/geOrga_/statuses/290596205795373057"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3

```

```

.. ..$ source      : chr "web"
.. ..$ type        : chr "twitter"
..$ klout           :List of 1
.. ..$ score: int 42
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag          : chr "en"
..$ salience       :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int -6
..$ twitter         :List of 5
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:29 +0000"
.. ..$ id          : chr "290596205795373057"
.. ..$ source      : chr "web"
.. ..$ text        : chr "so sick of how i look and what i dress like"
.. ..$ user        :List of 14
.. .. ..$ created_at      : chr "Sat, 13 Oct 2012 08:51:55 +0000"
.. .. ..$ description     : chr "Bring me the horizon, Batman, Red Hot Chili Peppers, Imogen's lettuce and Deez Nut"
.. .. ..$ followers_count : int 328
.. .. ..$ friends_count   : int 397
.. .. ..$ id              : int 877378586
.. .. ..$ id_str          : chr "877378586"
.. .. ..$ lang            : chr "en"
.. .. ..$ name            : chr "bruce wayne"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3104325484/a158d36a6aae44df6fe6ad791b9d7c47_normal.jpg"
.. .. ..$ screen_name     : chr "ge0rga_"
.. .. ..$ statuses_count  : int 3576
.. .. ..$ time_zone       : chr "Amsterdam"
.. .. ..$ url             : chr "http://suicidalsins.tumblr.com/"
.. .. ..$ utc_offset      : int 3600
$ :List of 5

```

```

..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/3065924952/4f74e4cfb76b27bab717869211ed9ebd_normal.jpeg"
.. .. ..$ id       : int 703476594
.. .. ..$ link     : chr "http://twitter.com/wishiwassgold"
.. .. ..$ name     : chr "wishiwassgold"
.. .. ..$ username: chr "wishiwassgold"
.. ..$ content     : chr "@DonnaNewcross I'd have to wear net knickers. At least plenty of supplies on ward, toiletries"
.. ..$ created_at : chr "Sun, 13 Jan 2013 23:08:34 +0000"
.. ..$ id         : chr "1e25dd627674a500e0749ff9146f4cc6"
.. ..$ link       : chr "http://twitter.com/wishiwassgold/statuses/290596229891637248"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Twitter for Android"
.. ..$ type       : chr "twitter"
..$ klout         :List of 1
.. ..$ score: int 45
..$ language     :List of 2
.. ..$ confidence: int 100
.. ..$ tag        : chr "en"
..$ salience    :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 8
..$ twitter     :List of 10
.. ..$ created_at : chr "Sun, 13 Jan 2013 23:08:34 +0000"
.. ..$ id         : chr "290596229891637248"
.. ..$ in_reply_to_screen_name: chr "DonnaNewcross"
.. ..$ in_reply_to_status_id : chr "290595430927040513"
.. ..$ in_reply_to_user_id   : chr "32157576"
.. ..$ mention_ids          :List of 1
.. .. ..$ : int 32157576

```

```

.. ..$ mentions           :List of 1
.. .. ..$ : chr "DonnaNewcross"
.. ..$ source              : chr "<a href=\"http://twitter.com/download/android\" rel=\"nofollow\">Twitter for Android"
.. ..$ text                : chr "@DonnaNewcross I'd have to wear net knickers. At least plenty of supplies on way home"
.. ..$ user                :List of 14
.. .. ..$ created_at       : chr "Wed, 18 Jul 2012 17:38:56 +0000"
.. .. ..$ description      : chr "#NHS Nurse, single parent, socialist (3 things Tories hate), avid reader (all genres)"
.. .. ..$ followers_count  : int 127
.. .. ..$ friends_count    : int 129
.. .. ..$ id               : int 703476594
.. .. ..$ id_str           : chr "703476594"
.. .. ..$ lang             : chr "en"
.. .. ..$ location         : chr "UK"
.. .. ..$ name             : chr "wishiwasgold"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3065924952/4f74e4cfb76b27bab717869211ed9ebd_normal.jpg"
.. .. ..$ screen_name      : chr "wishiwasgold"
.. .. ..$ statuses_count   : int 3466
.. .. ..$ time_zone        : chr "Amsterdam"
.. .. ..$ utc_offset       : int 3600
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "male"
..$ interaction:List of 8
.. ..$ author :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/2149169845/DSCN0674_normal.jpg"
.. .. ..$ id      : int 346929724
.. .. ..$ link    : chr "http://twitter.com/JLeonelFabulous"
.. .. ..$ name     : chr "Leonel Escobar"
.. .. ..$ username: chr "JLeonelFabulous"
.. ..$ content    : chr "Watching #thegoldenblobs #redcarpet"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:43 +0000"

```



```

.. ..$ id      : chr "1e25dd62cc48af80e0742d941b4fe788"
.. ..$ link     : chr "http://twitter.com/JLeoneIFabulous/statuses/290596267590049793"
.. ..$ schema   :List of 1
.. .. ..$ version: int 3
.. ..$ source   : chr "Twitter for iPhone"
.. ..$ type     : chr "twitter"
..$ klout       :List of 1
.. ..$ score: int 26
..$ language    :List of 2
.. ..$ confidence: int 63
.. ..$ tag       : chr "en"
..$ salience    :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter      :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:43 +0000"
.. ..$ hashtags  :List of 2
.. .. ..$ : chr "thegoldenblobs"
.. .. ..$ : chr "redcarpet"
.. ..$ id        : chr "290596267590049793"
.. ..$ source    : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. ..$ text      : chr "Watching #thegoldenblobs #redcarpet"
.. ..$ user      :List of 15
.. .. ..$ created_at      : chr "Tue, 02 Aug 2011 01:43:50 +0000"
.. .. ..$ description     : chr "Im just the representation of artistic and creative expression on earth... Conquer
.. .. ..$ followers_count : int 125
.. .. ..$ friends_count  : int 684
.. .. ..$ id              : int 346929724
.. .. ..$ id_str          : chr "346929724"
.. .. ..$ lang            : chr "en"
.. .. ..$ location        : chr "Your imagination "

```

```

.. .. ..$ name           : chr "Leonel Escobar"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2149169845/DSCN0674_normal.jpg"
.. .. ..$ screen_name     : chr "JLeonelFabulous"
.. .. ..$ statuses_count  : int 1886
.. .. ..$ time_zone       : chr "Central America"
.. .. ..$ url             : chr "http://hausofj1.tumblr.com/"
.. .. ..$ utc_offset      : int -21600
$:List of 5
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/2359599606/pcfq2187y99qnrf7nh8l_normal.jpeg"
.. .. ..$ id       : int 19463303
.. .. ..$ link     : chr "http://twitter.com/WendyWings"
.. .. ..$ name     : chr "Wendy Lester"
.. .. ..$ username: chr "WendyWings"
.. ..$ content    : chr "#eredcarpet is the hashtag to use if you want to send comments to the social team"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:47 +0000"
.. ..$ id        : chr "1e25dd62f26ea980e0749b2db9dd9268"
.. ..$ link      : chr "http://twitter.com/WendyWings/statuses/290596281200566272"
.. ..$ schema    :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Twitter for Android"
.. ..$ type       : chr "twitter"
..$ klout        :List of 1
.. ..$ score: int 59
..$ language     :List of 2
.. ..$ confidence: int 100
.. ..$ tag       : chr "en"
..$ salience    :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0

```

```

..$ twitter      :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:47 +0000"
.. ..$ hashtags  :List of 1
.. .. ..$ : chr "eredcarpet"
.. ..$ id        : chr "290596281200566272"
.. ..$ source    : chr "<a href=\"http://twitter.com/download/android\" rel=\"nofollow\">Twitter for Android</a>"
.. ..$ text      : chr "#eredcarpet is the hashtag to use if you want to send comments to the social team"
.. ..$ user      :List of 15
.. .. ..$ created_at      : chr "Sat, 24 Jan 2009 21:05:45 +0000"
.. .. ..$ description     : chr "Wife,mother,Narnian.I am on the list. All round good egg according to the NZ Herald"
.. .. ..$ followers_count : int 5185
.. .. ..$ friends_count   : int 1945
.. .. ..$ id              : int 19463303
.. .. ..$ id_str          : chr "19463303"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 240
.. .. ..$ location        : chr "Auckland, New Zealand"
.. .. ..$ name            : chr "Wendy Lester"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2359599606/pcfq2187y99qnrf7nh8l_normal.jpeg"
.. .. ..$ screen_name     : chr "WendyWings"
.. .. ..$ statuses_count  : int 107638
.. .. ..$ time_zone       : chr "Auckland"
.. .. ..$ utc_offset      : int 43200
$ :List of 5
..$ demographic:List of 1
.. ..$ gender: chr "mostly_female"
..$ interaction:List of 8
.. ..$ author   :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/2924151879/6820bcb53b0c8b84e7b05f3209fb24da_normal.jpeg"
.. .. ..$ id     : int 39005026
.. .. ..$ link   : chr "http://twitter.com/lauren_albrecht"

```

```

.. .. ..$ name      : chr "Lauren "
.. .. ..$ username: chr "lauren_albrecht"
.. ..$ content      : chr "Its my favorite time if year, red carpet season!! #GoldenGlobes #Eredcarpet"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:47 +0000"
.. ..$ id           : chr "1e25dd62f26ea980e074c0e67366d0e2"
.. ..$ link         : chr "http://twitter.com/lauren_albrecht/statuses/290596284660842496"
.. ..$ schema       :List of 1
.. .. ..$ version: int 3
.. ..$ source       : chr "Twitter for iPhone"
.. ..$ type         : chr "twitter"
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag         : chr "en"
..$ salience      :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 12
..$ twitter        :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:47 +0000"
.. ..$ hashtags    :List of 2
.. .. ..$ : chr "GoldenGlobes"
.. .. ..$ : chr "Eredcarpet"
.. ..$ id          : chr "290596284660842496"
.. ..$ source      : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. ..$ text        : chr "Its my favorite time if year, red carpet season!! #GoldenGlobes #Eredcarpet"
.. ..$ user        :List of 16
.. .. ..$ created_at      : chr "Sun, 10 May 2009 05:45:54 +0000"
.. .. ..$ description     : chr "play the best memories, pause the unforgettable moments, stop the heartbreaks, rew
.. .. ..$ followers_count : int 235
.. .. ..$ friends_count  : int 382
.. .. ..$ geo_enabled     : logi TRUE
.. .. ..$ id              : int 39005026

```

```

.. .. ..$ id_str          : chr "39005026"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 1
.. .. ..$ name            : chr "Lauren "
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2924151879/6820bcb53b0c8b84e7b05f3209fb24da_normal.jpg"
.. .. ..$ screen_name     : chr "lauren_albrecht"
.. .. ..$ statuses_count  : int 4061
.. .. ..$ time_zone       : chr "Central Time (US & Canada)"
.. .. ..$ url             : chr "http://www.facebook.com/#!/lealbrecht"
.. .. ..$ utc_offset      : int -21600
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "male"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/2937055250/f1960dc2b25c2b216e698dfccd09f41d_normal.jpeg"
.. .. ..$ id       : int 534689273
.. .. ..$ link     : chr "http://twitter.com/rafaeltcardoza"
.. .. ..$ name     : chr "Rafael T. Cardoza"
.. .. ..$ username: chr "rafaeltcardoza"
.. ..$ content    : chr "Red Carpet #GoldenGlobes"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:53 +0000"
.. ..$ id        : chr "1e25dd632ba7a080e0748f4a1b9add2a"
.. ..$ link      : chr "http://twitter.com/rafaeltcardoza/statuses/290596307809218560"
.. ..$ schema    :List of 1
.. .. ..$ version: int 3
.. ..$ source    : chr "Twitter for BlackBerry@"
.. ..$ type      : chr "twitter"
..$ klout       :List of 1
.. ..$ score: int 37
..$ language    :List of 2

```

```

.. ..$ confidence: int 100
.. ..$ tag      : chr "en"
..$ salience   :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter     :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:53 +0000"
.. ..$ hashtags  :List of 1
.. .. ..$ : chr "GoldenGlobes"
.. ..$ id        : chr "290596307809218560"
.. ..$ source    : chr "<a href=\"http://blackberry.com/twitter\" rel=\"nofollow\">Twitter for BlackBerry®</a>"
.. ..$ text      : chr "Red Carpet #GoldenGlobes"
.. ..$ user      :List of 12
.. .. ..$ created_at      : chr "Fri, 23 Mar 2012 20:22:57 +0000"
.. .. ..$ description     : chr "En los caminos que Dios me ponga!!"
.. .. ..$ followers_count : int 92
.. .. ..$ friends_count  : int 79
.. .. ..$ id             : int 534689273
.. .. ..$ id_str         : chr "534689273"
.. .. ..$ lang           : chr "es"
.. .. ..$ location       : chr "Valencia - Venezuela"
.. .. ..$ name           : chr "Rafael T. Cardoza"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2937055250/f1960dc2b25c2b216e698dfccd09f41d_normal.jpg"
.. .. ..$ screen_name     : chr "rafaeltcardoza"
.. .. ..$ statuses_count  : int 1130
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "female"
..$ interaction:List of 8
.. ..$ author   :List of 5
.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/2950563857/7664af9b9c9cfc7e2f75d0457ab6d757_normal.jpeg"

```

```

.. ..$ id      : int 519089925
.. ..$ link     : chr "http://twitter.com/frafra12"
.. ..$ name     : chr "franca"
.. ..$ username: chr "frafra12"
.. ..$ content  : chr "@GiulianaRancic love your second outfit. You look absolutely stunning!!!"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:58 +0000"
.. ..$ id       : chr "1e25dd635b56a100e074a5f2ed2e4190"
.. ..$ link     : chr "http://twitter.com/frafra12/statuses/290596328524890113"
.. ..$ schema   :List of 1
.. ..$ version: int 3
.. ..$ source   : chr "Twitter for Android"
.. ..$ type     : chr "twitter"
..$ klout       :List of 1
.. ..$ score: int 17
..$ language    :List of 2
.. ..$ confidence: int 100
.. ..$ tag       : chr "en"
..$ salience   :List of 1
.. ..$ content:List of 1
.. ..$ sentiment: int 7
..$ twitter    :List of 9
.. ..$ created_at      : chr "Sun, 13 Jan 2013 23:08:58 +0000"
.. ..$ id              : chr "290596328524890113"
.. ..$ in_reply_to_screen_name: chr "GiulianaRancic"
.. ..$ in_reply_to_user_id   : chr "21681809"
.. ..$ mention_ids          :List of 1
.. ..$ : int 21681809
.. ..$ mentions             :List of 1
.. ..$ : chr "GiulianaRancic"
.. ..$ source               : chr "<a href=\"http://twitter.com/download/android\" rel=\"nofollow\">Twitter for Ar
.. ..$ text                 : chr "@GiulianaRancic love your second outfit. You look absolutely stunning!!!"

```

```

.. ..$ user                :List of 11
.. .. ..$ created_at       : chr "Fri, 09 Mar 2012 02:10:42 +0000"
.. .. ..$ followers_count  : int 8
.. .. ..$ friends_count    : int 54
.. .. ..$ id               : int 519089925
.. .. ..$ id_str           : chr "519089925"
.. .. ..$ lang             : chr "en"
.. .. ..$ location         : chr "alliston"
.. .. ..$ name             : chr "franca"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2950563857/7664af9b9c9cfc7e2f75d0457ab6d757_normal.jpg"
.. .. ..$ screen_name      : chr "frafra12"
.. .. ..$ statuses_count   : int 42
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "female"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/2963559504/0643b40d7cebccef188a06999c884cb1_normal.jpeg"
.. .. ..$ id       : int 17902637
.. .. ..$ link     : chr "http://twitter.com/amberjfinney"
.. .. ..$ name     : chr "Amber Jené"
.. .. ..$ username: chr "amberjfinney"
.. ..$ content     : chr "RT @_IFB: Best description ever RT @fuggirls I really cannot with the color of Kelly's dress"
.. ..$ created_at  : chr "Sun, 13 Jan 2013 23:08:59 +0000"
.. ..$ id         : chr "1e25dd6364dfa780e074831b640f422c"
.. ..$ link       : chr "http://twitter.com/amberjfinney/statuses/290596331230224384"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Twitter for iPhone"
.. ..$ type       : chr "twitter"
..$ klout        :List of 1

```



```

.. ..$ score: int 55
..$ language :List of 2
.. ..$ confidence: int 100
.. ..$ tag : chr "en"
..$ salience :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter :List of 3
.. ..$ id : chr "290596331230224384"
.. ..$ retweet :List of 9
.. .. ..$ count : int 1
.. .. ..$ created_at : chr "Sun, 13 Jan 2013 23:08:59 +0000"
.. .. ..$ hashtags :List of 1
.. .. .. ..$ : chr "goldenglobes"
.. .. ..$ id : chr "290596331230224384"
.. .. ..$ mention_ids:List of 1
.. .. .. ..$ : int 21279256
.. .. ..$ mentions :List of 1
.. .. .. ..$ : chr "fuggirls"
.. .. ..$ source : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. .. ..$ text : chr "Best description ever RT @fuggirls I really cannot with the color of Kelly's dress. It's
.. .. ..$ user :List of 17
.. .. .. ..$ created_at : chr "Fri, 05 Dec 2008 17:35:50 +0000"
.. .. .. ..$ description : chr "tiramisu?"
.. .. .. ..$ followers_count : int 1886
.. .. .. ..$ friends_count : int 984
.. .. .. ..$ geo_enabled : logi TRUE
.. .. .. ..$ id : int 17902637
.. .. .. ..$ id_str : chr "17902637"
.. .. .. ..$ lang : chr "en"
.. .. .. ..$ listed_count : int 56

```

```

.. .. ..$ location      : chr "nyc."
.. .. ..$ name          : chr "Amber Jené"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2963559504/0643b40d7cebccef188a06999c884cb1"
.. .. ..$ screen_name    : chr "amberjfinney"
.. .. ..$ statuses_count : int 139973
.. .. ..$ time_zone      : chr "Eastern Time (US & Canada)"
.. .. ..$ url            : chr "http://ambersuntitledblog.com"
.. .. ..$ utc_offset     : int -18000
.. ..$ retweeted:List of 4
.. .. ..$ created_at: chr "Sun, 13 Jan 2013 23:08:42 +0000"
.. .. ..$ id         : chr "290596260367437825"
.. .. ..$ source      : chr "<a href=\"http://www.hootsuite.com\" rel=\"nofollow\">HootSuite</a>"
.. .. ..$ user        :List of 16
.. .. .. ..$ created_at      : chr "Tue, 06 Apr 2010 21:31:49 +0000"
.. .. .. ..$ description     : chr "Independent Fashion Bloggers is a collective mentorship dedicated to helping bl
.. .. .. ..$ followers_count : int 35482
.. .. .. ..$ friends_count   : int 3900
.. .. .. ..$ id              : int 130281555
.. .. .. ..$ id_str          : chr "130281555"
.. .. .. ..$ lang            : chr "en"
.. .. .. ..$ listed_count    : int 1153
.. .. .. ..$ location        : chr "New York, NY "
.. .. .. ..$ name            : chr "IFB "
.. .. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2602743907/rgkahu1es9qad81jko1y_normal.png"
.. .. .. ..$ screen_name     : chr "_IFB"
.. .. .. ..$ statuses_count  : int 28742
.. .. .. ..$ time_zone       : chr "Pacific Time (US & Canada)"
.. .. .. ..$ url             : chr "http://heartIFB.com"
.. .. .. ..$ utc_offset      : int -28800
$ :List of 6
..$ demographic:List of 1

```

```

.. ..$ gender: chr "female"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/1383294611/twitter_icon_normal.JPG"
.. .. ..$ id       : int 27067775
.. .. ..$ link     : chr "http://twitter.com/cinefille"
.. .. ..$ name     : chr "Joanna"
.. .. ..$ username: chr "cinefille"
.. ..$ content    : chr "My cat has stopped suffocating me. #GoldenGlobes red carpet time."
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:03 +0000"
.. ..$ id        : chr "1e25dd638b05a180e0748eda42c05a5e"
.. ..$ link      : chr "http://twitter.com/cinefille/statuses/290596349404131328"
.. ..$ schema    :List of 1
.. .. ..$ version: int 3
.. ..$ source    : chr "web"
.. ..$ type      : chr "twitter"
..$ klout       :List of 1
.. ..$ score: int 47
..$ language    :List of 2
.. ..$ confidence: int 100
.. ..$ tag       : chr "en"
..$ salience   :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 6
..$ twitter     :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:03 +0000"
.. ..$ hashtags  :List of 1
.. .. ..$ : chr "GoldenGlobes"
.. ..$ id        : chr "290596349404131328"
.. ..$ source    : chr "web"
.. ..$ text      : chr "My cat has stopped suffocating me. #GoldenGlobes red carpet time."

```

```

.. ..$ user      :List of 17
.. .. ..$ created_at      : chr "Fri, 27 Mar 2009 18:46:44 +0000"
.. .. ..$ description     : chr "On Tuesdays I run a classic film screening and get heckled by octogenarians. I blo
.. .. ..$ followers_count : int 764
.. .. ..$ friends_count   : int 651
.. .. ..$ geo_enabled     : logi TRUE
.. .. ..$ id              : int 27067775
.. .. ..$ id_str          : chr "27067775"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 39
.. .. ..$ location        : chr "New Jersey"
.. .. ..$ name            : chr "Joanna"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/1383294611/twitter_icon_normal.JPG"
.. .. ..$ screen_name     : chr "cinefille"
.. .. ..$ statuses_count  : int 14802
.. .. ..$ time_zone       : chr "Eastern Time (US & Canada)"
.. .. ..$ url             : chr "http://www.cine-fille.com"
.. .. ..$ utc_offset      : int -18000
$ :List of 5
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/3051798300/122c3df90e85e1692e95af53f7268675_normal.jpeg"
.. .. ..$ id       : int 24466642
.. .. ..$ link     : chr "http://twitter.com/stary1680"
.. .. ..$ name     : chr "Oriena "
.. .. ..$ username: chr "stary1680"
.. ..$ content     : chr "Holy OMG Zooey Deschanel looks Fabulous!!!! AHHH the red is the new trend calling it #Eredca
.. ..$ created_at  : chr "Sun, 13 Jan 2013 23:09:05 +0000"
.. ..$ id         : chr "1e25dd639e17ae80e074331d959ddbcb"
.. ..$ link       : chr "http://twitter.com/stary1680/statuses/290596360107999233"
.. ..$ schema     :List of 1

```

```

.. .. ..$ version: int 3
.. ..$ source      : chr "Echofon"
.. ..$ type        : chr "twitter"
..$ klout           :List of 1
.. ..$ score: int 39
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag          : chr "en"
..$ salience       :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 4
..$ twitter         :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:05 +0000"
.. ..$ hashtags    :List of 2
.. .. ..$ : chr "Eredcarpet"
.. .. ..$ : chr "GoldenGlobes"
.. ..$ id           : chr "290596360107999233"
.. ..$ source       : chr "<a href=\"http://www.echofon.com/\" rel=\"nofollow\">Echofon</a>"
.. ..$ text         : chr "Holy OMG Zooey Deschanel looks Fabulous!!!! AHHH the red is the new trend calling it #Eredca
.. ..$ user         :List of 16
.. .. ..$ created_at      : chr "Sun, 15 Mar 2009 01:39:38 +0000"
.. .. ..$ description     : chr "Comedic. Gleek! HIMYM! Pop-Culture lover. bookish. Canadian! Fanatic. Musicals. as
.. .. ..$ followers_count : int 1313
.. .. ..$ friends_count  : int 959
.. .. ..$ id              : int 24466642
.. .. ..$ id_str          : chr "24466642"
.. .. ..$ lang            : chr "en"
.. .. ..$ listed_count    : int 216
.. .. ..$ location        : chr "Toronto, CA"
.. .. ..$ name            : chr "Oriena "
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3051798300/122c3df90e85e1692e95af53f7268675_nor

```

```

.. .. ..$ screen_name      : chr "stary1680"
.. .. ..$ statuses_count   : int 24554
.. .. ..$ time_zone        : chr "Eastern Time (US & Canada)"
.. .. ..$ url               : chr "http://awhimsicalchatterbox.blogspot.com/"
.. .. ..$ utc_offset        : int -18000
$ :List of 5
..$ demographic:List of 1
.. ..$ gender: chr "male"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/3078355950/5aefdd453797a352cf612d026f30980c_normal.jpeg"
.. .. ..$ id        : int 623784093
.. .. ..$ link       : chr "http://twitter.com/sophieamos644"
.. .. ..$ name       : chr "Aslan"
.. .. ..$ username   : chr "sophieamos644"
.. ..$ content      : chr "RT @OfficialMerveeG: seeing girls still wearing Pauls Boutique.. http://t.co/Wfw6rreG"
.. ..$ created_at   : chr "Sun, 13 Jan 2013 23:09:10 +0000"
.. ..$ id           : chr "1e25dd63cdc6af00e0744e827875abca"
.. ..$ link         : chr "http://twitter.com/sophieamos644/statuses/290596379137568768"
.. ..$ schema       :List of 1
.. .. ..$ version: int 3
.. ..$ source       : chr "Twitter for iPhone"
.. ..$ type         : chr "twitter"
..$ language       :List of 2
.. ..$ confidence: int 100
.. ..$ tag         : chr "en"
..$ salience      :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter       :List of 3
.. ..$ id         : chr "290596379137568768"

```

```

.. ..$ retweet :List of 7
.. .. ..$ count      : int 28
.. .. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:10 +0000"
.. .. ..$ id         : chr "290596379137568768"
.. .. ..$ media      :List of 1
.. .. .. ..$ :List of 9
.. .. .. .. ..$ type      : chr "photo"
.. .. .. .. ..$ url       : chr "http://t.co/Wfw6rreG"
.. .. .. .. ..$ media_url  : chr "http://pbs.twimg.com/media/BAhmCLgCMAMbyLw.jpg"
.. .. .. .. ..$ display_url : chr "pic.twitter.com/Wfw6rreG"
.. .. .. .. ..$ expanded_url : chr "http://twitter.com/OfficialMerveeG/status/290594363594125312/photo/1"
.. .. .. .. ..$ id        : num 2.91e+17
.. .. .. .. ..$ media_url_https: chr "https://pbs.twimg.com/media/BAhmCLgCMAMbyLw.jpg"
.. .. .. .. ..$ sizes      :List of 4
.. .. .. .. .. ..$ medium:List of 3
.. .. .. .. .. .. ..$ h      : int 480
.. .. .. .. .. .. ..$ resize: chr "fit"
.. .. .. .. .. .. ..$ w      : int 320
.. .. .. .. .. .. ..$ large :List of 3
.. .. .. .. .. .. ..$ h      : int 480
.. .. .. .. .. .. ..$ resize: chr "fit"
.. .. .. .. .. .. ..$ w      : int 320
.. .. .. .. .. .. ..$ small :List of 3
.. .. .. .. .. .. ..$ h      : int 480
.. .. .. .. .. .. ..$ resize: chr "fit"
.. .. .. .. .. .. ..$ w      : int 320
.. .. .. .. .. .. ..$ thumb :List of 3
.. .. .. .. .. .. ..$ h      : int 150
.. .. .. .. .. .. ..$ resize: chr "crop"
.. .. .. .. .. .. ..$ w      : int 150
.. .. .. .. .. ..$ id_str    : chr "290594363598319619"

```

```

.. .. ..$ source      : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. .. ..$ text        : chr "seeing girls still wearing Pauls Boutique.. http://t.co/Wfw6rreG"
.. .. ..$ user        :List of 11
.. .. .. ..$ created_at      : chr "Sun, 01 Jul 2012 12:30:50 +0000"
.. .. .. ..$ description    : chr "Follow this girl@LouisaRossGower she's my bestfriend+absolutely gorgeous I foll
.. .. .. ..$ followers_count : int 852
.. .. .. ..$ friends_count  : int 1274
.. .. .. ..$ id             : int 623784093
.. .. .. ..$ id_str         : chr "623784093"
.. .. .. ..$ lang           : chr "en"
.. .. .. ..$ name           : chr "Aslan"
.. .. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/3078355950/5aefdd453797a352cf612d026f30980c_
.. .. .. ..$ screen_name     : chr "sophieamos644"
.. .. .. ..$ statuses_count  : int 267
.. ..$ retweeted:List of 4
.. .. ..$ created_at: chr "Sun, 13 Jan 2013 23:01:10 +0000"
.. .. ..$ id        : chr "290594363594125312"
.. .. ..$ source     : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. .. ..$ user       :List of 16
.. .. .. ..$ created_at      : chr "Fri, 22 Oct 2010 15:50:59 +0000"
.. .. .. ..$ description    : chr "Facebook: http://www.facebook.com/OfficialMerveeG.. .. Instagram: OfficialMervee
.. .. .. ..$ followers_count : int 34738
.. .. .. ..$ friends_count  : int 25823
.. .. .. ..$ id             : int 206254898
.. .. .. ..$ id_str         : chr "206254898"
.. .. .. ..$ lang           : chr "en"
.. .. .. ..$ listed_count   : int 11
.. .. .. ..$ location       : chr "England KENT"
.. .. .. ..$ name           : chr "†"
.. .. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/2971796405/0053ec06caea088262a2eae80b467f33_
.. .. .. ..$ screen_name     : chr "OfficialMerveeG"

```



```

.. .. ..$ statuses_count    : int 7543
.. .. ..$ time_zone         : chr "Alaska"
.. .. ..$ url                : chr "http://www.facebook.com/OfficialMerveeGrr"
.. .. ..$ utc_offset        : int -32400
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "female"
..$ interaction:List of 8
.. ..$ author      :List of 5
.. .. ..$ avatar   : chr "http://a0.twimg.com/profile_images/1578380461/flynn_s_normal.jpg"
.. .. ..$ id       : int 236905823
.. .. ..$ link     : chr "http://twitter.com/MACrowley"
.. .. ..$ name     : chr "Maren Crowley"
.. .. ..$ username: chr "MACrowley"
.. ..$ content     : chr "Multitasking hard. Watching @online #GoldenGlobeAwards #ERedCarpet and catching up on #Back
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:12 +0000"
.. ..$ id         : chr "1e25dd63e0d9ac00e0742db47a24ebe4"
.. ..$ link       : chr "http://twitter.com/MACrowley/statuses/290596388746715137"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "web"
.. ..$ type       : chr "twitter"
..$ klout         :List of 1
.. ..$ score: int 40
..$ language      :List of 2
.. ..$ confidence: int 100
.. ..$ tag        : chr "en"
..$ salience     :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter      :List of 8

```

```

.. ..$ created_at : chr "Sun, 13 Jan 2013 23:09:12 +0000"
.. ..$ hashtags :List of 3
.. .. ..$ : chr "GoldenGlobeAwards"
.. .. ..$ : chr "ERedCarpet"
.. .. ..$ : chr "Bachelor"
.. ..$ id : chr "290596388746715137"
.. ..$ mention_ids:List of 1
.. .. ..$ : int 2883841
.. ..$ mentions :List of 1
.. .. ..$ : chr "eonline"
.. ..$ source : chr "web"
.. ..$ text : chr "Multitasking hard. Watching @eonline #GoldenGlobeAwards #ERedCarpet and catching up on #Bac
.. ..$ user :List of 13
.. .. ..$ created_at : chr "Tue, 11 Jan 2011 16:51:01 +0000"
.. .. ..$ followers_count : int 261
.. .. ..$ friends_count : int 492
.. .. ..$ id : int 236905823
.. .. ..$ id_str : chr "236905823"
.. .. ..$ lang : chr "en"
.. .. ..$ location : chr "Long Island, NY"
.. .. ..$ name : chr "Maren Crowley"
.. .. ..$ profile_image_url: chr "http://a0.twimg.com/profile_images/1578380461/flynn_s_normal.jpg"
.. .. ..$ screen_name : chr "MACrowley"
.. .. ..$ statuses_count : int 5659
.. .. ..$ time_zone : chr "Eastern Time (US & Canada)"
.. .. ..$ utc_offset : int -18000
$ :List of 6
..$ demographic:List of 1
.. ..$ gender: chr "mostly_female"
..$ interaction:List of 8
.. ..$ author :List of 5

```

```

.. .. ..$ avatar : chr "http://a0.twimg.com/profile_images/2925087111/be30e521f9094325dab03551cdc67c3b_normal.jpeg"
.. .. ..$ id      : int 573265648
.. .. ..$ link    : chr "http://twitter.com/LDeWoody"
.. .. ..$ name     : chr "Lauren Marie"
.. .. ..$ username: chr "LDeWoody"
.. ..$ content    : chr "#ERedCarpet Jennifer Lawrence &lt;333"
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:12 +0000"
.. ..$ id         : chr "1e25dd63e0d9ac00e074591d84d8000e"
.. ..$ link       : chr "http://twitter.com/LDeWoody/statuses/290596386909614080"
.. ..$ schema     :List of 1
.. .. ..$ version: int 3
.. ..$ source     : chr "Twitter for iPhone"
.. ..$ type       : chr "twitter"
..$ klout        :List of 1
.. ..$ score: int 26
..$ language     :List of 2
.. ..$ confidence: int 65
.. ..$ tag        : chr "en"
..$ salience    :List of 1
.. ..$ content:List of 1
.. .. ..$ sentiment: int 0
..$ twitter     :List of 6
.. ..$ created_at: chr "Sun, 13 Jan 2013 23:09:12 +0000"
.. ..$ hashtags  :List of 1
.. .. ..$ : chr "ERedCarpet"
.. ..$ id        : chr "290596386909614080"
.. ..$ source    : chr "<a href=\"http://twitter.com/download/iphone\" rel=\"nofollow\">Twitter for iPhone</a>"
.. ..$ text      : chr "#ERedCarpet Jennifer Lawrence &lt;333"
.. ..$ user      :List of 13
.. .. ..$ created_at : chr "Mon, 07 May 2012 03:34:31 +0000"
.. .. ..$ description : chr " Treat me like a princess, I'll treat you like a prince. Treat me like a game, I'll

```

```

.. .. .$ followers_count : int 63
.. .. .$ friends_count   : int 337
.. .. .$ id               : int 573265648
.. .. .$ id_str           : chr "573265648"
.. .. .$ lang             : chr "en"
.. .. .$ name             : chr "Lauren Marie"
.. .. .$ profile_image_url: chr "http://a0.twimg.com/profile_images/2925087111/be30e521f9094325dab03551cdc67c3b_nor
.. .. .$ screen_name      : chr "LDeWoody"
.. .. .$ statuses_count   : int 815
.. .. .$ time_zone        : chr "Eastern Time (US & Canada)"
.. .. .$ utc_offset       : int -18000

```

finally!

```
length(junk$interactions)
```

```
[1] 18
```

```
junk$interactions[[18]]
```

```

$demographic
$demographic$gender
[1] "mostly_female"

```

```

$interaction
$interaction$author
$interaction$author$avatar
[1] "http://a0.twimg.com/profile_images/2925087111/be30e521f9094325dab03551cdc67c3b_normal.jpeg"

```

```
$interaction$author$id
```

```
[1] 573265648
```

```
$interaction$author$link
```

```
[1] "http://twitter.com/LDeWoody"
```

```
$interaction$author$name
```

```
[1] "Lauren Marie"
```

```
$interaction$author$username
```

```
[1] "LDeWoody"
```

```
$interaction$content
```

```
[1] "#ERedCarpet Jennifer Lawrence &lt;333"
```

```
$interaction$created_at
```

```
[1] "Sun, 13 Jan 2013 23:09:12 +0000"
```

```
$interaction$id
```

```
[1] "1e25dd63e0d9ac00e074591d84d8000e"
```

```
$interaction$link
```

```
[1] "http://twitter.com/LDeWoody/statuses/290596386909614080"
```

```
$interaction$schema
```

```
$interaction$schema$version
```

```
[1] 3
```

```
$interaction$source  
[1] "Twitter for iPhone"
```

```
$interaction$type  
[1] "twitter"
```

```
$klout  
$klout$score  
[1] 26
```

```
$language  
$language$confidence  
[1] 65
```

```
$language$tag  
[1] "en"
```

```
$salience  
$salience$content  
$salience$content$sentiment  
[1] 0
```

```
$twitter  
$twitter$created_at  
[1] "Sun, 13 Jan 2013 23:09:12 +0000"
```

\$twitter\$hashtags

\$twitter\$hashtags[[1]]

[1] "ERedCarpet"

\$twitter\$id

[1] "290596386909614080"

\$twitter\$source

[1] "Twitter for iPhone"

\$twitter\$text

[1] "#ERedCarpet Jennifer Lawrence <333"

\$twitter\$user

\$twitter\$user\$created_at

[1] "Mon, 07 May 2012 03:34:31 +0000"

\$twitter\$user\$description

[1] " Treat me like a princess, I'll treat you like a prince. Treat me like a game, I'll show you how its played. "

\$twitter\$user\$followers_count

[1] 63

\$twitter\$user\$friends_count

[1] 337

\$twitter\$user\$id

[1] 573265648

\$twitter\$user\$id_str

```
[1] "573265648"
```

```
$twitter$user$lang
```

```
[1] "en"
```

```
$twitter$user$name
```

```
[1] "Lauren Marie"
```

```
$twitter$user$profile_image_url
```

```
[1] "http://a0.twimg.com/profile_images/2925087111/be30e521f9094325dab03551cdc67c3b_normal.jpeg"
```

```
$twitter$user$screen_name
```

```
[1] "LDeWoody"
```

```
$twitter$user$statuses_count
```

```
[1] 815
```

```
$twitter$user$time_zone
```

```
[1] "Eastern Time (US & Canada)"
```

```
$twitter$user$utc_offset
```

```
[1] -18000
```

Activity #3 Attendance

Please Greet Your Neighbor(s)

- Practice in a low stakes environment. So it becomes natural.
- Reminder: You need at least one different neighbor each week for your attendance credit (only one photo required)

- Reminder: You cannot receive credit if it's only you in the photo, please refrain from photoshopping/editing photos
- Reminder: You need to upload the photo for credit.
- Please try to greet & meet a new classmate, take a photograph.

Even more data sources

[NOAA](#)

[International Disaster Database](#)

[FEMA](#)

[World Happiness Report](#)

[UN Development Reports](#)

[Youth Behavioral Risk - CDC](#)

[Opportunity Insights](#)

Relational Operators

Braun & Murdoch Chapter 2.4.1

- Recall the Earthquake data. Suppose you were told to create a subset of data where the magnitude (MAG) must be at least 2.5 and the quality (QUALITY) must be “A”.
- How would you write the request?

```
# this works but is horribly inefficient
# you don't want to be occupying so much of your resources

MAG25 <- EQ2025[EQ2025$MAG >= 2.5, ]
ANDQUALITYA <- MAG25[MAG25$QUALITY == "A",]
```

- Let's spend a little time with the Boolean Algebra of R
- Wikipedia tells us: Introduced by George Boole, it uses operators like AND, OR, and NOT to model, analyze, and simplify logical expressions, forming the fundamental basis of digital circuit design and computer programming

A Truth Table

- The body of this table shows how logical operations behave for every possible combination of TRUE and FALSE.

Table 2.1. <i>Truth table for Boolean operations</i>						
Boolean R	A A	B B	not A !A	not B !B	A and B A & B	A or B A B
	TRUE	TRUE	FALSE	FALSE	TRUE	TRUE
	TRUE	FALSE	FALSE	TRUE	FALSE	TRUE
	FALSE	TRUE	TRUE	FALSE	FALSE	TRUE
	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE

Figure 1: Chapter 2.4.1 Truth Table

Row 1 – A = TRUE, B = TRUE.

Results:

not A → FALSE

not B → FALSE

A and B → TRUE (both are TRUE)

A or B → TRUE (at least one is TRUE)

Short version

- NOT (!) flips TRUE FALSE
- AND (&) TRUE only when both are TRUE
- OR (|) TRUE when **at least one** is TRUE

Example

```
AMAG <- EQ2025[ c(39, 1746, 2, 372), "MAG"]  
BQUAL <- EQ2025[ c(39, 1746, 2, 372), "QUALITY"]  
AMAG
```

```
[1] 2.61 4.30 1.21 1.23
```

```
BQUAL
```

```
[1] "A" "D" "A" "C"
```

apply R's version

```
!(AMAG >= 2.5)
```

```
[1] FALSE FALSE TRUE TRUE
```

```
(BQUAL != "A")
```

```
[1] FALSE TRUE FALSE TRUE
```

```
(AMAG >= 2.5) & (BQUAL == "A")
```

```
[1] TRUE FALSE FALSE FALSE
```

```
(AMAG >= 2.5) | (BQUAL == "A")
```

```
[1] TRUE TRUE TRUE FALSE
```

The && and ||

- Not tested on this, I don't ask you to demonstrate its use. But need to know for 102

The operators && and || are similar to & and |, but behave differently in two respects. First, they are *not* vectorized: only one calculation is done. Secondly, they are guaranteed to be evaluated from left to right, with the right-hand operand only evaluated if necessary. For example, if A is FALSE, then A && B will be FALSE regardless of the value of B, so B needn't be evaluated. This can save time if evaluating B would be very slow, and may make calculations easier, for example if evaluating B would cause an error when A was FALSE. This behavior is sometimes called *short-circuit evaluation*.

Figure 2: && and ||

Summary of Relational Operations

- From Braun & Murdoch

- Type

```
a <- c(3, 6, 9)
```

- To test which elements are greater than 4, type

```
a > 4
```

- To test which elements are exactly equal⁶ to 4, type

```
a == 4
```

- To test which elements are greater than or equal to 4, type

```
a >= 4
```

- To test which elements are not equal to 4, type

```
a != 4
```

- To print the elements of a which are greater than 4, type

```
a[a > 4]
```

- Type

```
b <- c(4, 6, 8)
```

- To test which elements of a are less than the corresponding elements of b, type

```
a < b
```

- To print the elements of a that are less than the corresponding elements of b, type

```
a[a < b]
```

⁶ Be careful with tests of equality. Because R works with only a limited number of decimal places rounding error can accumulate, and you may find surprising results, such as `49 * (4 / 49)` not being equal to 4.

Figure 3: Relational Operator

Factors

Factors

- Factors appear to be strings (text data), but are not:

```
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 ...
```

```
head(iris)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

- Species in `iris` looks like a string but isn't

Factors - What they do

- Factors are variables in R which take on a limited number of different values, typically these variables represent non-numerical or “categorical” information.
- We use factors in statistical modeling because non-numerical variables (e.g., genre — rap, rock, pop, electronic) enter into statistical models differently than numerical variables (one-hot encoding, indicator, dummy coding)
- Designating non-numerical data as factors insures that the more advanced model estimation functions (101ABC) in R will treat these types of variables correctly.
- Factors will also let us order a “categorical” value in any way we wish, not only alphabetically. This is also useful when modeling.

Example 1: importing csv

- If you were to point and click your way to an import of a .csv file, you might see the following.

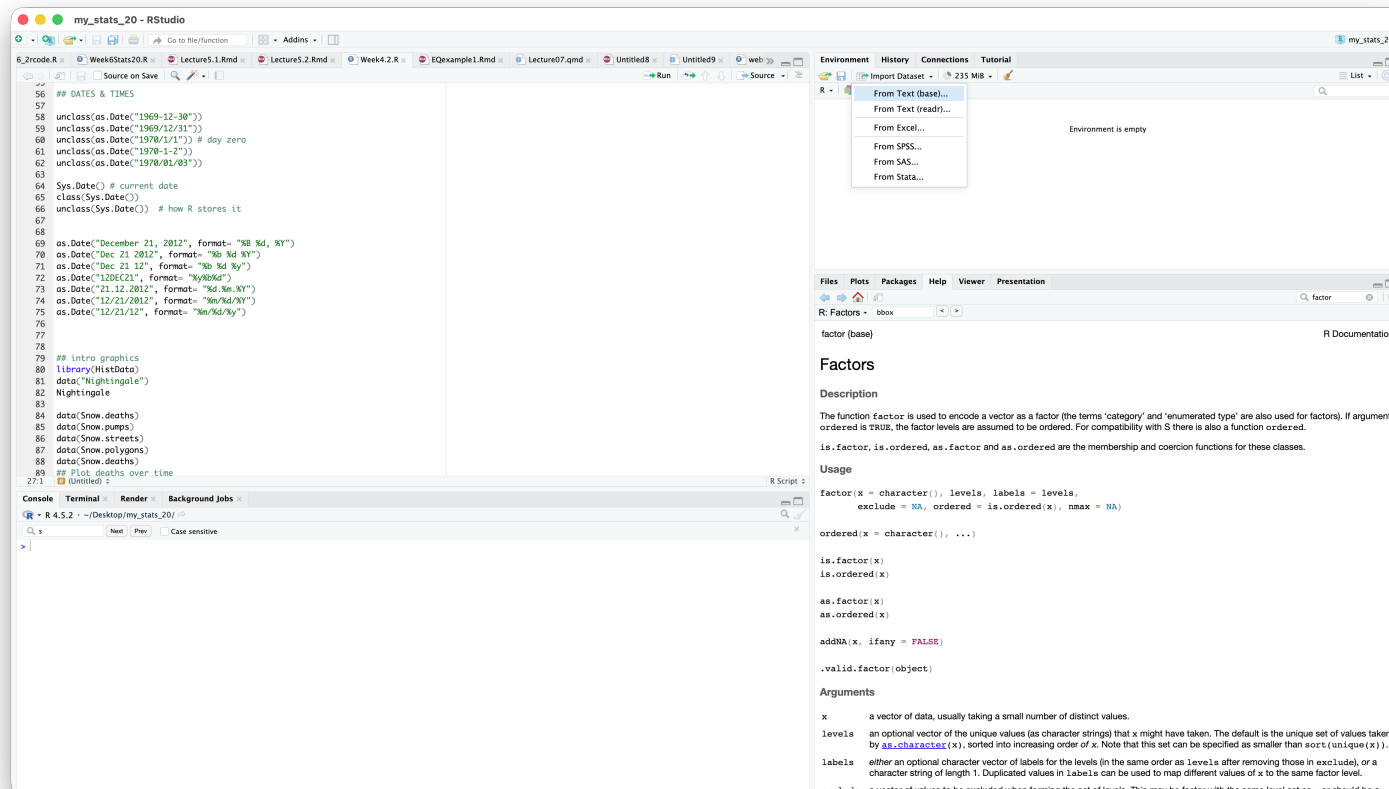


Figure 4: import

Example 1: import menu

- Factor is a choice which results in coercion on a string (character) variable

The screenshot shows the RStudio interface with the 'Import Dataset' dialog box open. The dialog box has the following fields and options:

- Name:** AAPL
- Input File:** Date, Open, High, Low, Close, Adj. Close, Volume
- Encoding:** Automatic
- Heading:** Yes (selected)
- Row names:** Automatic
- Separator:** Comma
- Decimal:** Period
- Quote:** Double (")
- Comment:** None
- na.strings:** NA
- ☐ Strings as factors

The 'Data Frame' preview shows the following data:

Date	Open	High	Low	Close	Adj. Close	Volume
Apr 27, 2026	266.09	268.36	265.07	267.61	267.61	41,213,866
Apr 24, 2026	272.76	273.06	269.65	271.06	271.06	38,124,500
Apr 23, 2026	275.05	275.77	271.65	273.43	273.43	33,399,600
Apr 22, 2026	267.82	273.74	266.87	273.17	273.17	43,249,200
Apr 21, 2026	271.5	272.8	265.4	266.17	266.17	50,209,800
Apr 20, 2026	270.33	274.28	270.29	273.05	273.05	36,590,200
Apr 17, 2026	266.96	272.3	266.72	270.23	270.23	61,436,200
Apr 16, 2026	266.8	267.16	261.27	263.4	263.4	43,323,100
Apr 15, 2026	258.16	266.56	257.81	266.43	266.43	49,913,500
Apr 14, 2026	259.25	261.93	257.19	258.83	258.83	48,370,700
Apr 13, 2026	259.73	260.18	256.66	259.2	259.2	36,234,700
Apr 10, 2026	259.98	262.19	259.02	260.48	260.48	31,291,500
Apr 9, 2026	259.00	261.12	256.07	260.49	260.49	22,121,600

The console shows the following R code:

```

56 ## DATES & TIMES
57
58 unclass(as.Date("1969-12-30"))
59 unclass(as.Date("1969/12/31"))
60 unclass(as.Date("1970/1/1")) # day zero
61 unclass(as.Date("1970-1-2"))
62 unclass(as.Date("1970/01/03"))
63
64 Sys.Date() # current date
65 class(Sys.Date())
66 unclass(Sys.Date()) # how R stores it
67
68
69 as.Date("December 21, 2012", format= "%B %d, %Y")
70 as.Date("Dec 21 2012", format= "%b %d %Y")
71 as.Date("Dec 21 12", format= "%b %d %y")
72 as.Date("12DEC21", format= "%Y%b%d")
73 as.Date("21.12.2012", format= "%d.%m.%Y")
74 as.Date("12/21/2012", format= "%m/%d/%Y")
75 as.Date("12/21/12", format= "%m/%d/%y")
76
77
78 ##
79 ## intro graphics
80 library(HistData)
81 data("Nightingale")
82 Nightingale
83
84 data$Snoo.deaths
85
27.1 (Untitled)

```

The environment pane shows 'my_stats_20' with 232 MB of memory. The history pane shows the 'Import Dataset' function call. The connections pane shows 'Global Environment'.

Figure 5: result

Example 1: strings as factor

The screenshot shows the RStudio interface with the 'Import Dataset' dialog box open. The dialog is configured for the 'AAPL' dataset, with the 'Input File' set to 'Date.Open.High.Low.Close.Adj.Close.Volume'. The 'Encoding' is set to 'Automatic', and the 'Strings as factors' checkbox is checked. The 'Data Frame' preview shows the following columns: Date, Open, High, Low, Close, Adj. Close, and Volume. The console shows the following R code:

```
## DATES & TIMES
56
57 unclass(as.Date("1969-12-30"))
58 unclass(as.Date("1969/12/31"))
59 unclass(as.Date("1970/1/1")) # day zero
60 unclass(as.Date("1970-1-2"))
61 unclass(as.Date("1970/01/03"))
62
63 Sys.Date() # current date
64 class(Sys.Date())
65 unclass(Sys.Date()) # how R stores it
66
67
68 as.Date("December 21, 2012", format= "%B %d, %Y")
69 as.Date("Dec 21 2012", format= "%b %d %Y")
70 as.Date("Dec 21 12", format= "%b %d %y")
71 as.Date("12DEC21", format= "%Y%b%d")
72 as.Date("21.12.2012", format= "%d.%m.%Y")
73 as.Date("12/21/2012", format= "%m/%d/%Y")
74 as.Date("12/21/12", format= "%m/%d/%y")
75
76
77
78 ## intro graphics
79 library(HistData)
80 data("Nightingale")
81
82
83 data("Snow.deaths")
84
```

The 'Console' shows the following output:

```
R 4.5.2 - ~/Desktop/my_stats_20/
>
```

The 'Environment' pane shows the 'Global Environment' with 232 MB of memory used. The 'History' pane shows the following commands:

```
1. Sys.Date()
2. class(Sys.Date())
3. unclass(Sys.Date())
4. as.Date("December 21, 2012", format= "%B %d, %Y")
5. as.Date("Dec 21 2012", format= "%b %d %Y")
6. as.Date("Dec 21 12", format= "%b %d %y")
7. as.Date("12DEC21", format= "%Y%b%d")
8. as.Date("21.12.2012", format= "%d.%m.%Y")
9. as.Date("12/21/2012", format= "%m/%d/%Y")
10. as.Date("12/21/12", format= "%m/%d/%y")
```

The 'Usage' pane shows the following code:

```
factor(x = character(), levels = levels,
       exclude = NA, ordered = is.ordered(x), nmax = NA)

ordered(x = character(), ...)

is.factor(x)
is.ordered(x)

as.factor(x)
```

Figure 6: stringsas

Example 1: result

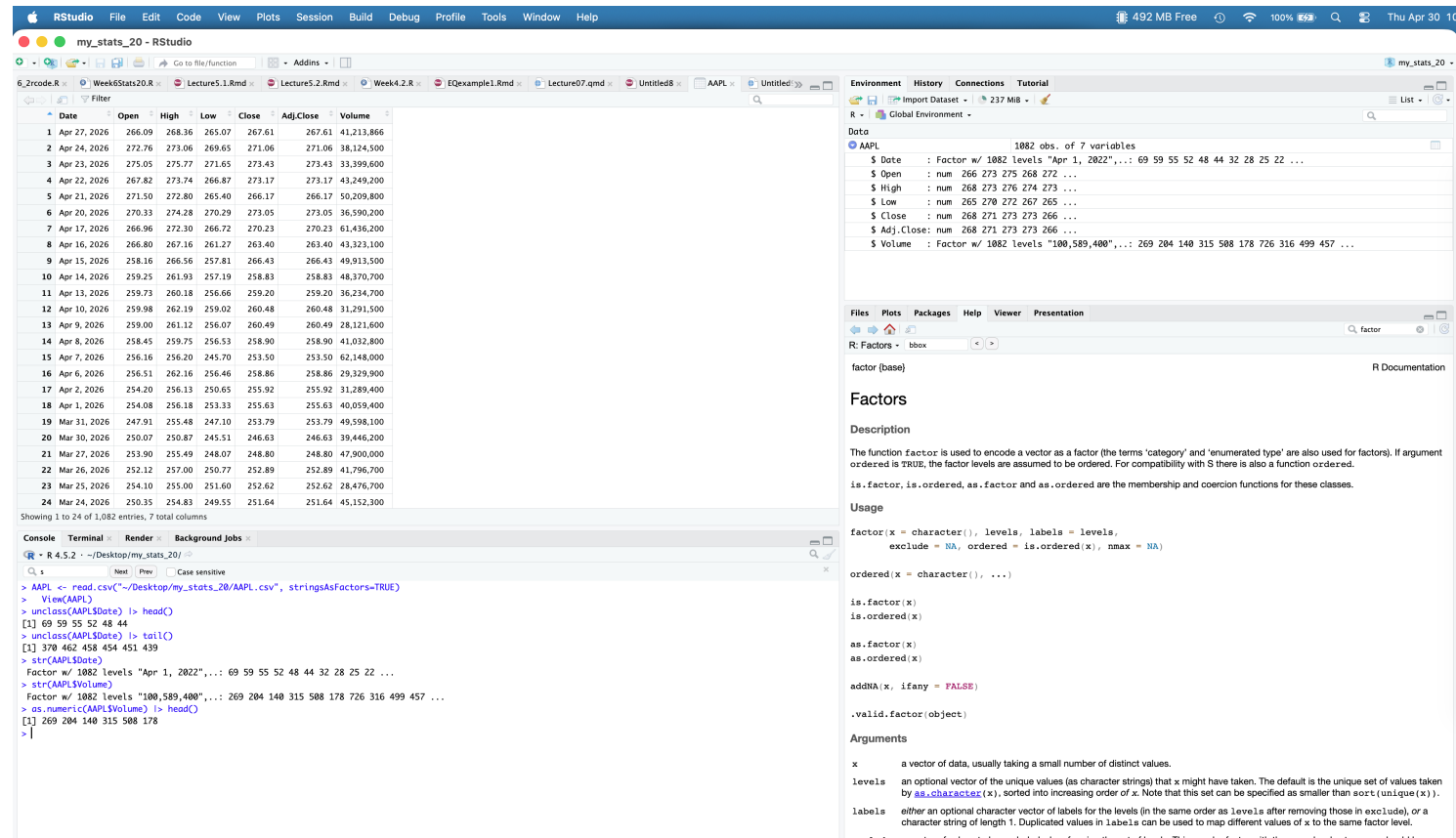


Figure 7: result

Example 2: another import

- Factors generate “ghost levels”

```
par(mfrow=c(2,1))

plot1 <- read.csv('http://www.stat.ucla.edu/~vlew/datasets/bom_imdb.csv')

plot1data = plot1[plot1$genre %in%
                  c("Action","Animation","Comedy","Crime","Drama"),]

boxplot(Budget ~ genre, data = plot1data,
        main='genre stored as character', xlab='Movie Genre',
        ylab= 'Budget in million dollars',
        col= c('red','green','blue','cyan','purple'), cex.lab=.65, cex.axis=.65)

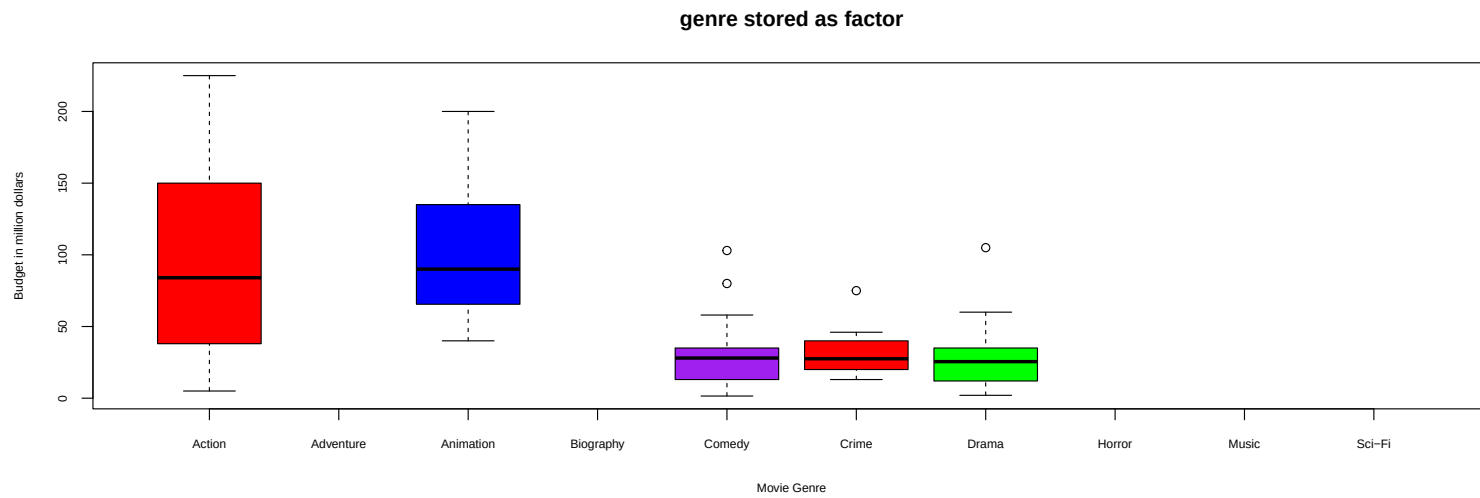
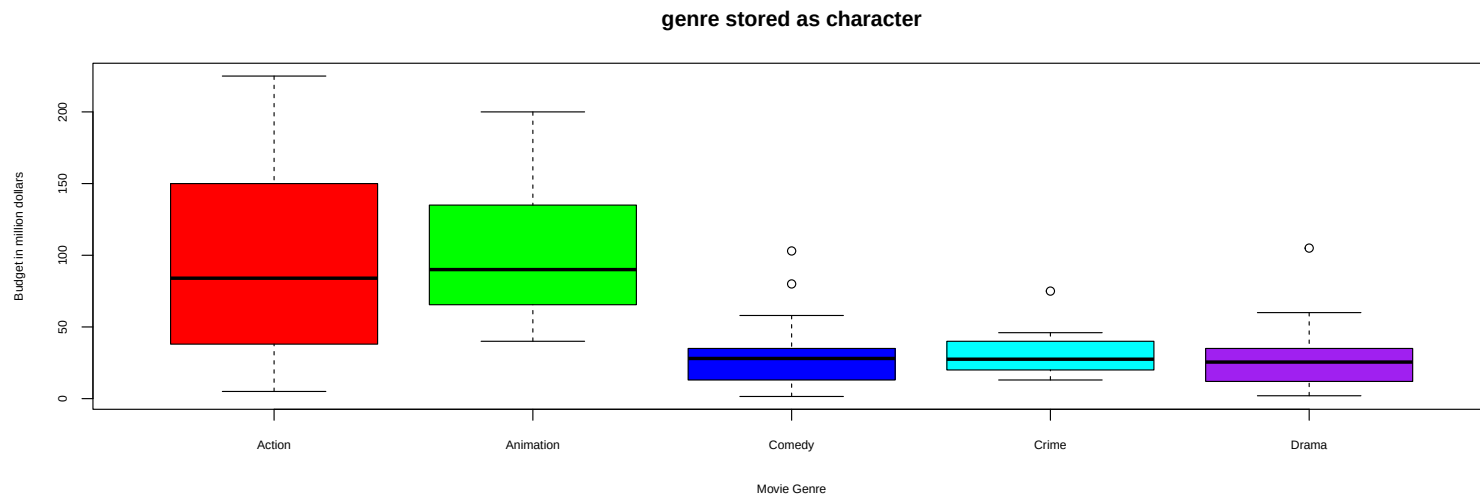
plot2 <- read.csv('http://www.stat.ucla.edu/~vlew/datasets/bom_imdb.csv',
                  stringsAsFactors=TRUE)

plot2data = plot2[plot2$genre %in%
                  c("Action","Animation","Comedy","Crime","Drama"),]

boxplot(Budget ~ genre, data= plot2data,
        main='genre stored as factor', xlab='Movie Genre',
        ylab= 'Budget in million dollars',
        col=c('red','green','blue','cyan','purple'), cex.lab=.65, cex.axis=.65)

par(mfrow=c(1,1))
```

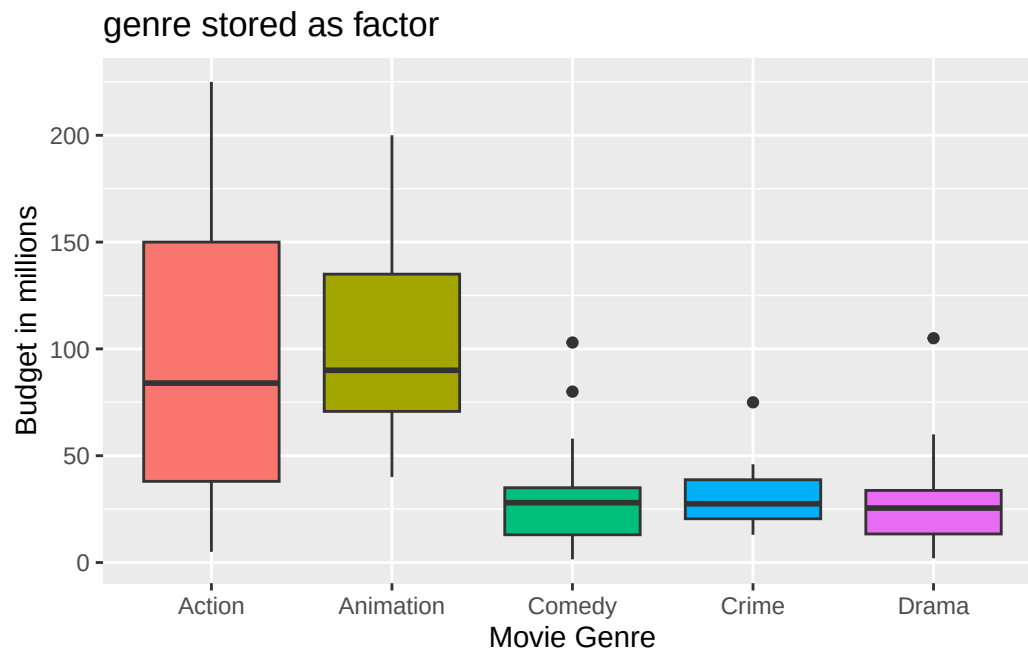
Result



Note about ggplot

```
library(ggplot2)
ggplot(plot2data, aes(x = genre, y = Budget, fill = genre)) +
  geom_boxplot() +
  labs(title = 'genre stored as factor', x = 'Movie Genre', y = 'Budget in millions') +
  theme(legend.position = "none")
```

Warning: Removed 54 rows containing non-finite outside the scale range
(`stat\_boxplot()`).



Thoughtless use of factors

- Read some data, the months are stored as strings, naturally

```
states <- read.csv(url("http://www.stat.ucla.edu/~vlew/datasets/state_covid_2020.csv"))  
  
str(states)
```

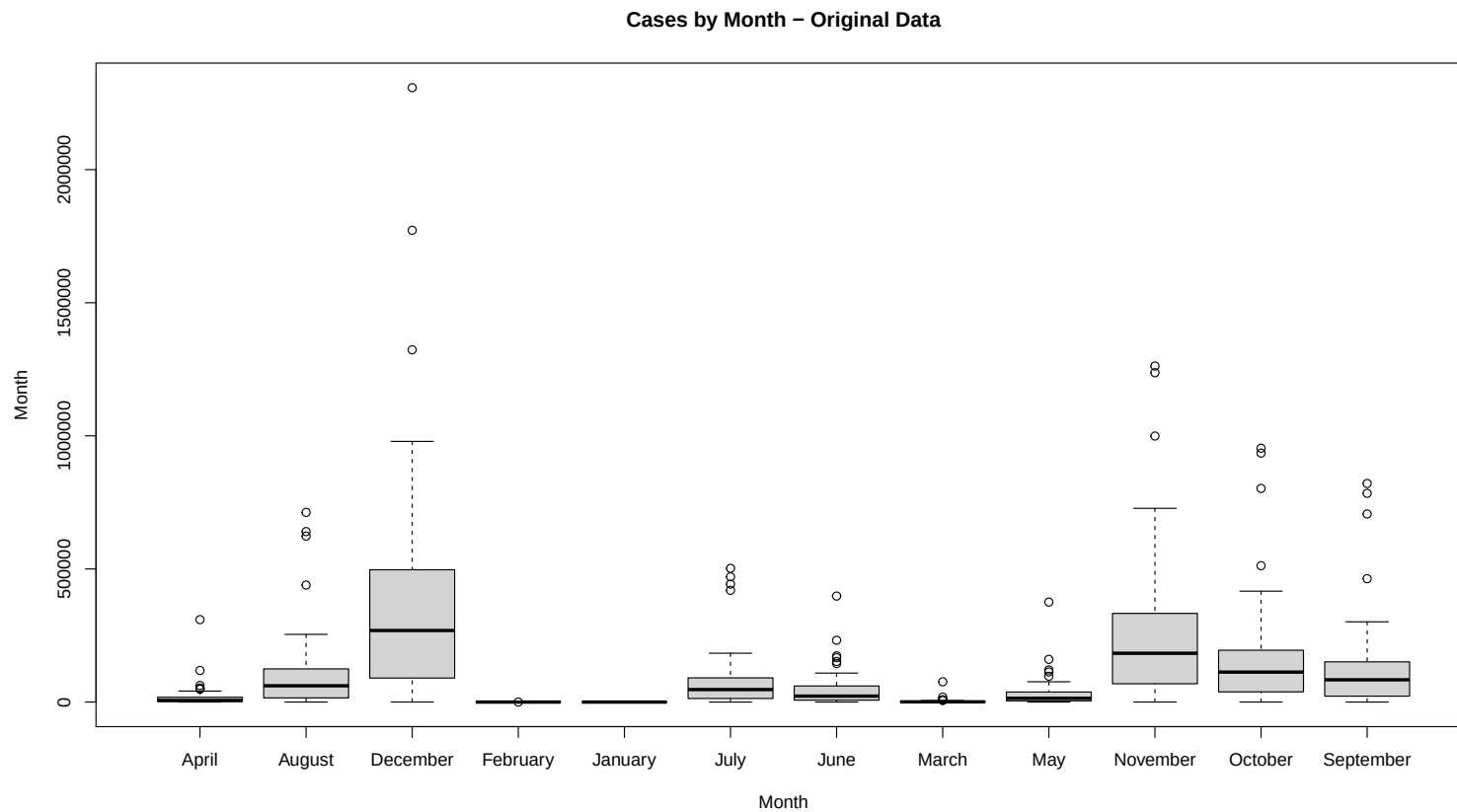
```
'data.frame':  564 obs. of  3 variables:  
 $ state: chr  "Alabama" "Alaska" "Arizona" "Arkansas" ...  
 $ MONTH: chr  "April" "April" "April" "April" ...  
 $ cases: int  7068 353 7648 3281 50470 15207 27700 4734 4323 33683 ...
```

- But do not do this to your audience

```
boxplot(cases ~ MONTH, data = states, main = "Cases by Month - Original Data",  
        xlab = "Month", ylab = "Month")
```

Thoughtless use of factors - result

- Strings are always alphabetized



A possible “fix”

```
states$MonthF <- factor(states$MONTH,  
                        levels=c("January", "February", "March", "April", "May", "June",  
                                "July", "August", "September", "October", "November", "December"))  
str(states)
```



```
'data.frame': 564 obs. of 4 variables:
 $ state : chr "Alabama" "Alaska" "Arizona" "Arkansas" ...
 $ MONTH : chr "April" "April" "April" "April" ...
 $ cases : int 7068 353 7648 3281 50470 15207 27700 4734 4323 33683 ...
 $ MonthF: Factor w/ 12 levels "January","February",...: 4 4 4 4 4 4 4 4 4 4 ...
```

```
head(states)
```

```
      state MONTH cases MonthF
1  Alabama April  7068  April
2   Alaska April   353  April
3  Arizona April  7648  April
4 Arkansas April  3281  April
5 California April 50470  April
6  Colorado April 15207  April
```

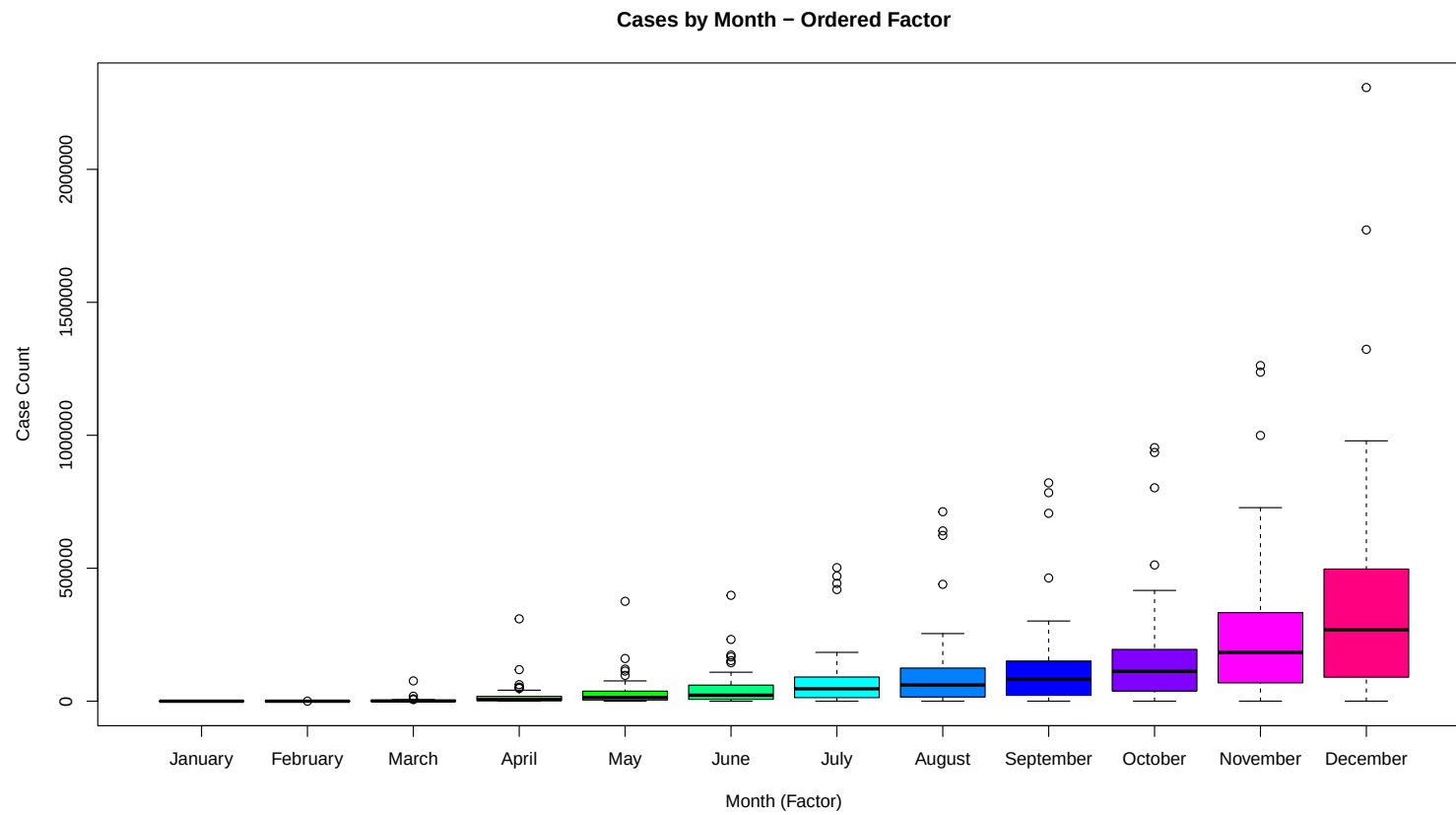
```
levels(states$MonthF)
```

```
[1] "January" "February" "March"    "April"    "May"      "June"
[7] "July"    "August"   "September" "October"  "November" "December"
```

- result on next slide

```
boxplot(cases ~ MonthF, data = states, main = "Cases by Month - Ordered Factor",
        xlab = "Month (Factor)", ylab = "Case Count", col = rainbow(12))
```

A possible “fix” - result



Dates and Times

Dates & Times

- Frequently ignored but ever present and often quite useful
- If not told otherwise, R will read data containing dates as we understand them as Character (or worse, Factor)
- Dates (there are exceptions) are stored by R as the number of days or seconds from January 1, 1970, using negative numbers for date before January 1, 1970.
- The default format expected by R is a four digit year, followed by a month, then a day, separated by either dashes or slashes.
- The base R `as.Date()` function converts character representations of dates (without times, e.g. 1/1/2020)
- The base R `strptime()` function converts character representations of dates and times to Unix (POSIX) time. This is used when data has times, time zones, AM/PM time stamps, etc.

Rules of Thumb

- The general rule for date/time data in R is to use the simplest technique possible.
- `as.Date()` will usually be the best choice because dates typically appear without times. Examples might be trading date, historical dates, birth date. The example from earlier on deaths in the workplace gives dates but not time.
- `strptime()` would usually be used if there is also a timestamp associated with the date. Generally, data generated by computer, such as a system date, will have a timestamp.
- there are additional base R functions such as
 - `as.POSIXlt()` - generates a LIST of components (good for extraction like day of the week).
 - `as.POSIXct()` - generates a numeric timestamp (good for computation)

Examples

The unclass function is only being used to reveal R's storage of the date (never will ask you about it):

```
unclass(as.Date("1969-12-30"))
```

```
[1] -2
```

```
unclass(as.Date("1969/12/31"))
```

```
[1] -1
```

```
unclass(as.Date("1970/1/1")) # day zero
```

```
[1] 0
```

```
unclass(as.Date("1970-1-2"))
```

```
[1] 1
```

```
unclass(as.Date("1970/01/03"))
```

```
[1] 2
```

```
Sys.Date() # current date
```

```
[1] "2026-04-30"
```

```
class(Sys.Date())
```

```
[1] "Date"
```

```
unclass(Sys.Date()) # how R stores it
```

```
[1] 20573
```

Activity #4 (if there is time)

- Try to calculate the number of days you've been alive
- I will select Mr. LeBron James of the Lakers - Dec 30, 1984
- There are other methods of formatting

```
days_alive <- Sys.Date() - as.Date("1984/12/30")

days_num <- as.numeric(days_alive)

paste(
  "Mr. LeBron James is",
  format(days_num, big.mark = ","), # format() will convert the number to a character
  "days old"
)
```

```
[1] "Mr. LeBron James is 15,096 days old"
```

```
# NOTE  
class(days_num)
```

```
[1] "numeric"
```

```
format(days_num, big.mark = ",") |> class()
```

```
[1] "character"
```

Other formats

- The `as.Date()` function allows a variety of input formats through its `format=` argument.
- R can handle virtually any reasonably formatted date by using the `format=`

```
as.Date("December 21, 2012", format= "%B %d, %Y")
```

```
[1] "2012-12-21"
```

```
as.Date("Dec 21 2012", format= "%b %d %Y")
```

```
[1] "2012-12-21"
```

```
as.Date("Dec 21 12", format= "%b %d %y")
```

```
[1] "2012-12-21"
```

```
as.Date("12DEC21", format= "%y%b%d")
```

```
[1] "2012-12-21"
```

```
as.Date("21.12.2012", format= "%d.%m.%Y")
```

```
[1] "2012-12-21"
```

```
as.Date("12/21/2012", format= "%m/%d/%Y")
```

```
[1] "2012-12-21"
```

```
as.Date("12/21/12", format= "%m/%d/%y")
```

```
[1] "2012-12-21"
```

Recall date/time formats

- try `help(strptime)`

Files
Plots
Packages
Help
Viewer
Presentation

factor

R: Date-time Conversion Functions to and from Character
bbox

The details of the formats are platform-specific, but the following are likely to be widely available: most are defined by the POSIX standard. A *conversion specification* is introduced by %, usually followed by a single letter or o or E and then a single letter. Any character in the format string not part of a conversion specification is interpreted literally (and %% gives %). Widely implemented conversion specifications include

%a

Abbreviated weekday name in the current locale on this platform. (Also matches full name on input: in some locales there are no abbreviations of names.) Case-insensitive on input.

%A

Full weekday name in the current locale. (Also matches abbreviated name on input.) Case-insensitive on input.

%b

Abbreviated month name in the current locale on this platform. (Also matches full name on input: in some locales there are no abbreviations of names.) Case-insensitive on input.

%B

Full month name in the current locale. (Also matches abbreviated name on input.) Case-insensitive on input.

%c

Date and time. Locale-specific on output, "%a %b %e %H:%M:%S %Y" on input.

%C

Century (00–99): the integer part of the year divided by 100.

%d

Day of the month as decimal number (01–31).

%D

Date format such as %m/%d/%y: the C99 standard says it should be that exact format (but not all OSes comply).

%e

Day of the month as decimal number (1–31), with a leading space for a single-digit number.

%F

Equivalent to %Y-%m-%d (the ISO 8601 date format).

%g

The last two digits of the week-based year (see %V). (Accepted but ignored on input.)

Time Example

- Recall the Earthquake Data

```
EQ2025$datetime_text <- paste(EQ2025$DATE, EQ2025$TIME)
```

```
EQ2025$timestamp_utc <- as.POSIXct(  
  EQ2025$datetime_text,  
  format = "%Y/%m/%d %H:%M:%OS",  
  tz = "UTC" # 8 hours ahead  
)
```

```
head(EQ2025[, c(1, 2, 14, 15)])
```

	DATE	TIME	datetime_text	timestamp_utc
1	2025/01/01	01:00:56.92	2025/01/01 01:00:56.92	2025-01-01 01:00:56
2	2025/01/01	01:16:28.99	2025/01/01 01:16:28.99	2025-01-01 01:16:28
3	2025/01/01	01:30:49.87	2025/01/01 01:30:49.87	2025-01-01 01:30:49
4	2025/01/01	01:54:51.66	2025/01/01 01:54:51.66	2025-01-01 01:54:51
5	2025/01/01	02:14:01.26	2025/01/01 02:14:01.26	2025-01-01 02:14:01
6	2025/01/01	02:30:22.12	2025/01/01 02:30:22.12	2025-01-01 02:30:22

Put it to use

- There is a library called “lubridate” (you are on your own), instead we can use Base R for most things, like finding the day of the week:

```
EQ2025$day_of_week <- weekdays(EQ2025$timestamp_utc) # this is a posixct object  
head(EQ2025[, c(1, 2, 14:16)])
```

	DATE	TIME	datetime_text	timestamp_utc	day_of_week
1	2025/01/01	01:00:56.92	2025/01/01 01:00:56.92	2025-01-01 01:00:56	Wednesday
2	2025/01/01	01:16:28.99	2025/01/01 01:16:28.99	2025-01-01 01:16:28	Wednesday
3	2025/01/01	01:30:49.87	2025/01/01 01:30:49.87	2025-01-01 01:30:49	Wednesday
4	2025/01/01	01:54:51.66	2025/01/01 01:54:51.66	2025-01-01 01:54:51	Wednesday
5	2025/01/01	02:14:01.26	2025/01/01 02:14:01.26	2025-01-01 02:14:01	Wednesday
6	2025/01/01	02:30:22.12	2025/01/01 02:30:22.12	2025-01-01 02:30:22	Wednesday

```
table(EQ2025$day_of_week) |> sort()
```

Saturday	Sunday	Thursday	Friday	Wednesday	Tuesday	Monday
2204	2321	2394	2409	2447	2452	2591

```
EQ2025$dowF <- factor(EQ2025$day_of_week,
                      levels=c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"))
head(EQ2025[, c(1, 2, 14:17)])
```

	DATE	TIME	datetime_text	timestamp_utc	day_of_week
1	2025/01/01	01:00:56.92	2025/01/01 01:00:56.92	2025-01-01 01:00:56	Wednesday
2	2025/01/01	01:16:28.99	2025/01/01 01:16:28.99	2025-01-01 01:16:28	Wednesday
3	2025/01/01	01:30:49.87	2025/01/01 01:30:49.87	2025-01-01 01:30:49	Wednesday
4	2025/01/01	01:54:51.66	2025/01/01 01:54:51.66	2025-01-01 01:54:51	Wednesday
5	2025/01/01	02:14:01.26	2025/01/01 02:14:01.26	2025-01-01 02:14:01	Wednesday
6	2025/01/01	02:30:22.12	2025/01/01 02:30:22.12	2025-01-01 02:30:22	Wednesday

dowF

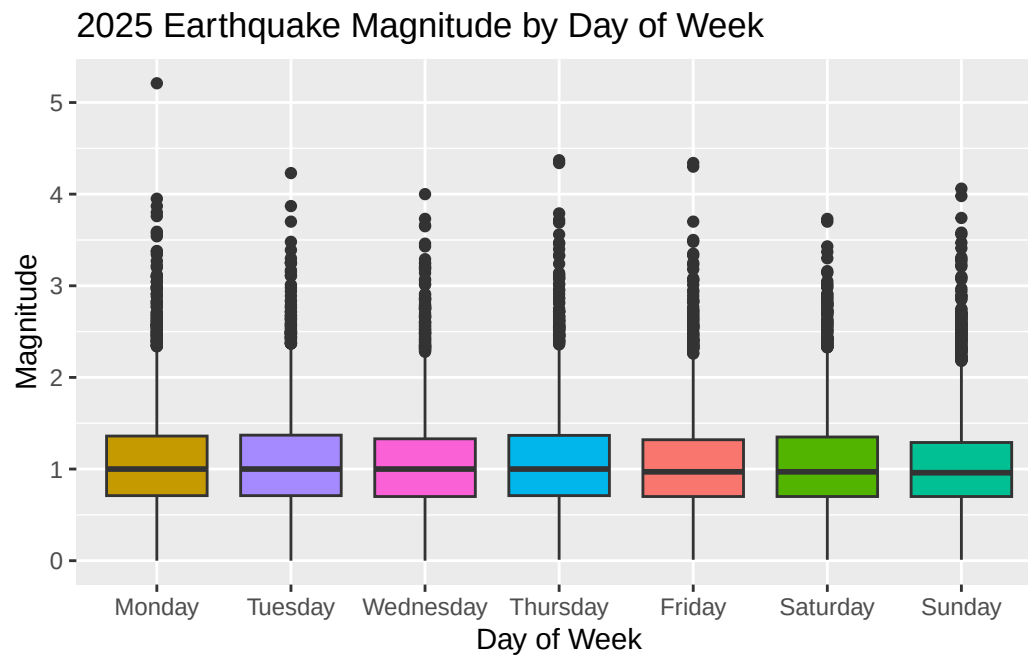
1	Wednesday
2	Wednesday
3	Wednesday
4	Wednesday

5 Wednesday

6 Wednesday

Put it to use 2

```
ggplot(data = EQ2025, aes(x = dowF, y = MAG, fill = day_of_week)) +  
  geom_boxplot() +  
  labs(title = "2025 Earthquake Magnitude by Day of Week", x = "Day of Week", y = "Magnitude") +  
  theme(legend.position = "none")
```



Have a terrific weekend